Periodic cyclic homology of smooth crossed product algebras

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November 2026

Abstract

This paper is devoted to the study of smooth convolution algebras associated to Lie groups, appearing naturally in non-commutative geometry and representation theory. Our result establishes a stable isomorphism between the periodic cyclic homology of the smooth crossed product algebras associated to a real Lie group and to a maximal compact subgroup of it. It may be viewed as an homological analogue to the Dirac induction with coefficients for Lie groups. Our approach is built on earlier work of Nistor who established this isomorphism locally, *i.e.* on each conjugacy class of the group and without stabilization. Essential tools come from Kasparov's bivariant K-theory and its interpretation by Cuntz. We set up a refinement of the Dirac-dual Dirac method and of Morita equivalence which fit into our framework.

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1 Introduction

Let G be an almost-connect real Lie group and let K be a maximal compact subgroup of it. The contractibility of the homogeneous space G/K provides one of the central bridges between representation theory, index theory, and non-commutative geometry. A recurring principle is that, although G might be analytically complicated, many of its invariants can be compared with invariants attached to the compact group K. The present work develops such a comparison in periodic cyclic homology for smooth crossed product algebras.

The starting point is the Connes-Kasparov theorem which provides an isomorphism up to a degree shift

$$K_\bullet(C_r^\star(K)) \xrightarrow{\sim} K_{\bullet+\dim(G/K)}(C_r^\star(G))$$

between the K-theoretical invariants of the associated reduced $C^\star$-algebras. This comparison is implemented by Dirac induction. The Dirac operator on the symmetric space G/K, together with its dual class, give rise to mutually inverse elements in equivariant Kasparov theory. Representation-theoretically, this result makes correspond the unitary dual of K and the tempered dual of G, as developed in the work of Wassermann and Mackey.

If a locally compact group G acts continuously on a $C^\star$-algebra A, we can construct a reduced crossed product $A \rtimes_r G$ which can be thought as the reduced $C^\star$-algebra of G with coefficients in A. It encodes both representations of the group G and modules over the algebra A. Naturally, one might consider a version of the Connes-Kasparov isomorphism with coefficients. It is the purpose of the Baum-Connes conjecture with coefficients for Lie groups, which remains unproven.

Conjecture 1. *Suppose an almost-connect real Lie group G with maximal compact subgroup K acts continuously on a $C^\star$-algebra A, then there exists a degree-shift isomorphism*

$$K_\bullet(A \rtimes_r K) \longrightarrow K_{\bullet+\dim(G/K)}(A \rtimes_r G). \tag{1}$$

Although K-groups are notoriously difficult to compute, the periodic cyclic homology provides an approximation which is more determinable. The reduced crossed product algebras being well-adapted to K-theory, their smooth counterparts called *smooth crossed products algebras* are especially suitable for periodic cyclic homology. These analogies suggest that the Connes-Kasparov isomorphism should have a cyclic-homological counterpart. This was explored by Nistor in the early 1990s and the main motivation for this paper.

Given a locally convex algebra A endowed with a smooth action of a real Lie group G, the associated **smooth** crossed product is the convolution algebra $A \rtimes G := \mathcal{C}_c^\infty(G, A)$. Nistor showed that in the case of a real algebraic group G with maximal compact subgroup K, the periodic cyclic homology of the algebras $A \rtimes G$ and $A \rtimes K$ are modules over the algebra of smooth class functions on G, and isomorphic after localization at maximal ideals of this algebra [Nis93].

Theorem 1.1 (Nistor)**.** *Let A be a locally convex algebra endowed with a smooth action of a real algebraic group G. For any maximal ideal $\mathfrak{m}$ of the smooth class functions algebra $\mathcal{C}^\infty(G)^{inv}$, there exists an isomorphism functorial in A*

$$HP_\bullet(A \rtimes K)_\mathfrak{m} \simeq HP_{\bullet+\dim(G/K)}(A \rtimes G)_\mathfrak{m}.$$

This result can be thought as a *local* homological-analogue of the Baum-Connes conjecture with coefficients for Lie groups. It is already a strong indication that the geometry of G/K governs the cyclic theory of smooth crossed products, just as it governs the K-theory of reduced crossed products. Nevertheless, the passage from local statements to a global isomorphism is subtle. The local description involves centralizers of elements of G and their geometry does not vary smoothly enough across conjugacy classes to permit a direct gluing argument. A global construction requires a mechanism that avoids decomposing the group into local contributions and instead works directly with the geometry of G/K.

The purpose of this paper is to provide such a global construction after a natural stabilization. Our main result establishes that for any locally convex unital algebra A endowed with a smooth action of an almost-connected real Lie group G with maximal compact subgroup K, there exists an isomorphism

$$HP_\bullet(\mathcal{A} \rtimes K) \xrightarrow{\sim} HP_{\bullet+\dim(G/K)}(\mathcal{A} \rtimes G) \tag{2}$$

after stabilization $\mathcal{A} = A \otimes_\pi \mathcal{D}$ by a certain G-smooth Schatten ideal $\mathcal{D}$ (see 4.12 for its definition). This statement may be regarded as an homological analogue of Dirac induction with coefficients. Two ingredients organize the proof.

The first is a smooth version of Green's imprimitivity theorem. In the locally convex setting, usual Morita equivalence is replaced by a weaker notion, called quasi-Morita equivalence. The Cuntz-Quillen description of periodic cyclic homology using quasi-free extensions and X-complexes shows that quasi-Morita equivalence yields continuous chain-homotopy equivalence in periodic cyclic homology. This paper develops a smooth-equivariant version of this equivalence according to [Cun98], compatible with smooth crossed products in periodic cyclic homology. Namely, if G is a real Lie group acting smooth over a locally convex algebra A, one has

$$A \sim_{G-\text{quasi-Morita}} B \;\Rightarrow\; \widehat{CC}(A \rtimes G) \xrightarrow{\sim} \widehat{CC}(B \rtimes G).$$

Later we shall prove that for any closed subgroup $H < G$, one has a G-equivariant quasi-Morita equivalent between the locally convex G-algebras

$$\mathcal{C}_c^\infty(G, A) \rtimes H \sim_{G-\text{quasi-Morita}} \mathcal{C}_c^\infty(G/H, A)$$

which, once applied twice, yields to the following isomorphism.

Theorem. *Let A be a locally convex unital algebra endowed with a smooth action of a real Lie group G and H any closed subgroup of G. There exists an isomorphism*

$$HP_\bullet(A \rtimes H) \xrightarrow{\sim} HP_\bullet(\mathcal{C}_c^\infty(G/H, A) \rtimes G). \tag{3}$$

The second ingredient is obtained by adapting Kasparov's Dirac-dual Dirac method to periodic cyclic homology. In this part, G is supposed to be an almost-connected real Lie group with maximal compact subgroup K. On the symmetric space G/K, Kasparov constructed a G-equivariant Dirac element α and a K-equivariant dual Dirac element β whose Kasparov products are inverse after restriction to the maximal compact subgroup K [Kas81]. This equivariant duality is the main argument behind the proof of the Connes-Kasparov theorem. To transport it to smooth crossed products, one needs an algebraic model of bivariant classes that is compatible with locally convex algebras and crossed products.
This paper provides such a model by developing an analogue of the Cuntz picture [Cun87] in this framework. Classically, a KK-class is represented by a $\star$-homomorphism out of the Cuntz algebra qA, and the Kasparov product corresponds to composition to the associated $\star$-homomorphisms after stabilization. Here we introduce a Fréchet version $q_R A$ of the Cuntz algebra for fixed parameter $R > 0$, defined as a completion of the algebraic free product with respect to a family of submultiplicative norms. This construction is functorial, stable under group actions, and compatible with crossed products. Within this setting, the Dirac and dual Dirac elements can be represented by equivariant homomorphisms of locally convex algebras and induce linear maps between the periodic cyclic complexes of the corresponding crossed products by the Cuntz-Quillen formalism

$$HP(\alpha \rtimes G) : HP_\bullet(\mathcal{C}_c^\infty(G/K, \mathcal{A}) \rtimes G) \longrightarrow HP_{\bullet+\dim(G/K)}(\mathcal{A} \rtimes G)$$

$$HP(\beta \rtimes K) : HP_\bullet(\mathcal{A} \rtimes K) \longrightarrow HP_{\bullet+\dim(G/K)}(\mathcal{C}_c^\infty(G/K, \mathcal{A}) \rtimes K)$$

where $\mathcal{A} = A \otimes_\pi \mathcal{D}$ denotes the stabilization of the algebra A with a G-smooth Schatten ideal $\mathcal{D}$. Indeed, the linear maps $HP(\alpha \rtimes K)$ and $HP(\beta \rtimes K)$ are not immediately inverse to each others on periodic cyclic homology and the stabilization algebra $\mathcal{D}$ is designed precisely to absorb this defect.
Finally, the crucial observation is that their K-equivariant duality suffices to produce a global isomorphism after passing to crossed products by the whole group G. Indeed, the Nistor's local comparison theorem 1.1 applied to the map $HP(\alpha \rtimes G)$ shows that it becomes an isomorphism on every stalk, namely a global isomorphism. In this way, the K-equivariant duality of Dirac and dual Dirac elements transported to periodic cyclic homology, yields the following degree-shift isomorphism.

Theorem. *Let A be locally convex algebra endowed with a smooth action of an almost-connect real Lie group G with maximal compact subgroup K. The Kasparov's Dirac element α induces an isomorphism*

$$HP(\alpha \rtimes G) : HP_\bullet(\mathcal{C}_c^\infty(G/K, \mathcal{A}) \rtimes G) \xrightarrow{\sim} HP_{\bullet+\dim(G/K)}(\mathcal{A} \rtimes G). \qquad (4)$$

after stabilization $\mathcal{A} = A \otimes_\pi \mathcal{D}$ by a certain G-smooth Schatten ideal $\mathcal{D}$.

Combining these two ingredients yields the isomorphism (2), stated in theorem 4.20.

Periodic cyclic homology

Let us breifly review the definition and main properties of periodic cyclic homology for the class of locally convex algebras. Fix A to be a locally convex algebra,

namely a topological complex algebra whose underlying vector space is locally convex an whose multiplication is separately continuous. For $n \geq 0$, we consider the vector space

$$C_n(A) := A^{\otimes_\pi n} \oplus A^{\otimes_\pi n+1}$$

and write

$$C_\star(A) = \bigoplus_{n\geq 0} C_n(A)$$

for the associated graded vector space. They are endowed with the Hochschild's and Connes' differentials [Lod92]

$$b : C_n(A) \longrightarrow C_{n-1}(A) \quad \text{and} \quad B : C_n(A) \longrightarrow C_{n+1}(A) \tag{5}$$

verifying the fundamental properties

$$b^2 = B^2 = (b+B)^2 = 0.$$

The operator $b + B$ sends any even (*resp.* odd) degree element to a sum of odd (*resp.* even) degree elements, thus we split the grading into *even* and *odd* parts

$$C_{\text{even}}(A) := \prod_{n\geq 0} C_{2n}(A) \quad \text{and} \quad C_{\text{odd}}(A) := \prod_{n\geq 0} C_{2n+1}(A).$$

Definition 1.2 We call **periodic cyclic complex** of a locally convex algebra A the following $\mathbb{Z}/2\mathbb{Z}$-graded complex

$$\widehat{CC}_\bullet(A) := \quad C_{\text{even}}(A) \underset{b+B}{\overset{b+B}{\rightleftarrows}} C_{\text{odd}}(A) \ .$$

Its **periodic cyclic homology** is the homology of its periodic cyclic complex

$$HP_\bullet(A) := H_\bullet(\widehat{CC}_\bullet(A)).$$

Theorem 1.3. *The periodic cyclic homology respects the following properties:*

- *(Morita invariance)* [LQ84] *For any locally convex algebra A and $r \geq 1$, the trace map on $r \times r$-matrices over A induces an isomorphism in periodic cyclic homology*
$$Tr : HP_\bullet(M_r(A)) \xrightarrow{\sim} HP_\bullet(A).$$
- *(Excision)* [CQ97] [Pus01] *Given any locally convex (non-necessarily unital) algebra A and two-sided ideal $I \subseteq A$ such that $\pi : A \to A/I$ is endowed with a bounded linear section, there exists a chain-homotopy equivalence*
$$\widehat{CC}(I) \xrightarrow{\sim} \ker\Big(\widehat{CC}(A) \xrightarrow{\pi} \widehat{CC}(A/I)\Big).$$
- *(Smooth homotopy invariance)* [Kas87] *If $f_0, f_1 : A \to B$ are continuous morphisms of locally convex algebras, homotopic via a smooth family of continuous morphisms $f_t : A \to B$, then $(f_0)^\bullet = (f_1)^\bullet : HP_\bullet(B) \longrightarrow HP_\bullet(A)$. Moreover, if M is a compact manifold, the antisymetrization map induces an isomorphism between periodic cyclic homology of its algebra of smooth functions $\mathcal{C}^\infty(M)$ and its De Rham cohomology groups*
$$HP_0(\mathcal{C}^\infty(M)) \simeq \prod_{n\geq 0} H^{2n}_{dR}(M;\mathbb{C}) \quad \text{and} \quad HP_1(\mathcal{C}^\infty(M)) \simeq \prod_{n\geq 0} H^{2n+1}_{dR}(M;\mathbb{C}).$$

Definition 1.4 Suppose G is a real Lie group acting smoothly on a locally convex algebra A. We defined the associated **smooth crossed product** as the locally convex convolution algebra

$$A \rtimes G := \mathcal{C}_c^\infty(G, A) \quad (f_1 \star f_2)(g) = \int_G f_1(s)\ s \cdot (f_2(s^{-1}g))ds \in A.$$

The aim of this paper is to understand the periodic cyclic homology of this algebra.

2 Morita equivalences of crossed product algebras

Let G be a real Lie group acting smoothly on a complete locally convex algebra A. In this section we shall establish the isomorphism (3) for any closed subgroup $H < G$. To do so, we first use the description of the periodic cyclic homology via quasi-free extensions by Cuntz and Quillen, reviewed in [CQ95a] and [CQ95b]. Then, we develop the notion of quasi-Morita equivalence initially defined by Cuntz for a discrete group action [Cun98] to the case of smooth action of Lie groups to fit our context. Together, they will finally lead to the proposition 2.12 which, applied twice, yields the expected isomorphism (3) stated in theorem 2.13.

2.1 The Cuntz-Quillen method

In the 90', Cuntz and Quillen established a description of the periodic cyclic homology using quasi-free extension of algebras [CQ95a] and [CQ95b]. This point of view enriches the structure of periodic cyclic homology with a *derived functor property*.

Definition 2.1 Given a locally convex vector complex space W, we denote its locally convex **tensor algebra** by

$$TW := W \oplus (W \otimes_\pi W) \oplus (W \otimes_\pi W \otimes_\pi W) \oplus \cdots$$

whose continuous multiplication is given by the tensor product. It respects the universal property that any continuous linear map $W \to A$ from W to a locally convex algebra A can be uniquely extended to a continuous algebra homomorphism $TW \to A$.

Definition 2.2 According to [CQ95a], we say that a locally convex algebra R is **quasi-free** if, for any nilpotent extension S of R, there is a continuous lifting homorphism $R \to S$. A **quasi-free extension** of a locally convex algebra A is an extension $A = R/I$ with R quasi-free.

The canonical example of quasi-free extension of a locally convex algebra A is its tensor algebra TA defined above. Indeed, TA is free (hence quasi-free) by universal property and the multiplication ideal $IA = \ker(m : TA \to A)$ yields an extension $A = TA/IA$.

Definition 2.3 The **X-complex** of a locally convex algebra R is the $\mathbb{Z}/2\mathbb{Z}$-graded complex

$$X(R) := \quad R \underset{b}{\overset{B}{\rightleftarrows}} C_1(R)/\overline{b(C_2(R))}\ .$$

where b and B denote the Hochschild's and Connes' differentials as in (5).

Theorem 2.4 (Cuntz-Quillen [CQ95a])**.** *Fix R a quasi-free extension of a locally convex algebra $A = R/I$. The X-complex of the I-adic completion $\widehat{R} := \varprojlim R/I^n$ is naturally and continuously chain-homotopy equivalent to the periodic cyclic complex of A*

$$X(\widehat{R}) \xrightarrow{\sim} \widehat{CC}(A).$$

Corollary 2.5. *Suppose A is a locally convex algebra and W is locally convex vector space endowed with a continuous linear epimorphism $p : W \twoheadrightarrow A$ and a bounded linear section $s : A \to W$. There exists a continuous chain homotopy equivalence*

$$X(\widehat{TW}) \xrightarrow{\sim} \widehat{CC}(A)$$

where $\widehat{TW}$ denotes the completion of the tensor algebra TW with respect to the ideal $I = \ker(m \circ Tp : TW \to A)$.

Proof. Since the complete locally convex tensor algebra TA is a quasi-free extension of A, the theorem above tells us that there exists a natural and continuous chain-homotopy equivalence $X(\widehat{TA}) \xrightarrow{\sim} \widehat{CC}(A)$ where $\widehat{TA}$ denotes the adic completion of TA with respect to the ideal $IA = \ker(m : TA \to A)$. The statement will follow from the fact that the X-complexes of $\widehat{TW}$ and $\widehat{TA}$ are continuously chain-homotopy equivalent. We write $V := \ker(W \to A)$, so we get an exact sequence of vector spaces

$$0 \longrightarrow V \xrightarrow{i} W \xrightarrow{p} A \longrightarrow 0 \ ,$$

with bounded linear section $s : A \to W$. We shall prove that the morphism of algebras $Tp : TW \to TA$ is a continuous isomorphism of filtrated algebras, of inverse $Ts : TA \to TW$ modulo homotopy.
First, we show their continuity. By assumption, p and s are continuous for the locally convex topologies and by the universal property of the tensor algebra, the algebra homomorphisms Tp and Ts are also. We show that they also are continuous for the adic-topologies, namely that Tp and Ts are morphisms of filtrated algebras. If $x \in I \subseteq TW$ we get $m(Tp(x)) = (m \circ Tp)(x) = 0$, *i.e.* $Tp(x) \in IA$. On the other side, if $a \in IA \subseteq TA$, $(m \circ Tp)(Ts(a)) = (m \circ T(p \circ s))(a) = m(a) = 0$ which is that $Ts(a) \in I$.
Now, we show that Tp and Ts are inverse to each others modulo homotopy. By definition of s and p, it is clear that $Tp \circ Ts = id_{TA}$. Also, $(Ts \circ Tp)(Ti \star Ts) = T(s \circ p \circ i) \star T(s \circ p \circ s) = 0 \star Ts = (Ti \star Ts)(0 \star id)$ and the morphism $Ts \circ Tp$ fits into the following commutative diagram:

$$\begin{array}{ccc} TV \star TA & \xrightarrow{Ti \star Ts} & TW \\ {\scriptstyle 0 \star id}\downarrow & & \downarrow{\scriptstyle Ts \circ Tp} \\ TV \star TA & \xrightarrow{Ti \star Ts} & TW \end{array}$$

Consider the family of linear maps $h_t : V \to V, v \mapsto tv$ for $t \in [0, 1]$ and thus the morphisms $T(h_t) \star id : TV \star TA \to TV \star TA$. It is clear that the homotopy preserves the filtrations and that it is smooth with respect to the parameter $t \in [0, 1]$. Since it realizes an homotopy between $0 \star id$ and the identity of $TV \star TA$, it realizes through $Ti \star Ts$ an homotopy between $Ts \circ Tp$ and the identity of TW. The

fact that Tp is a continuous isomorphism modulo homotopy follows from the fact that the free homomorphism $Ti \star Ts$ is a continuous isomorphism. To show this free homomorphism is a continuous isomorphism, we construct a continuous inverse of it. The continuous linear map $\phi : w \in W \longmapsto (w - (s \circ p)(w)) + p(w) \in TV \star TA$ extends to a continuous morphism of algebras $id \star Tp : TW \longrightarrow TV \star TA$ by universal property of tensor algebras. Since $Tp \circ Ts = id$, it is clear that $(id \star Tp) \circ (Ti \star Ts) = id_{TV \star TA}$. On the other side

$$((Ti \star Ts) \circ \phi)(w) = (Ti \star Ts)((w - (s \circ p)(w)) + p(w)) = i(w - (s \circ p)(w)) + (s \circ p)(w) = w,$$

which implies that $(Ti \star Ts) \circ (id \star Tp) = id_{TW}$. Then $Ti \star Ts$ is a isomorphism and it ends up the proof. □

2.2 Quasi-Morita equivalence

In this section, we review the notion of quasi-Morita equivalence of algebras which is weaker than the notion of Morita equivalence, but compatible with periodic cyclic homology [Cun98]. Then we develop a smooth-equivariant version of it, defined in 2.9, in a way so that it fits the framework of smooth crossed product algebras as stated in corollary 2.11.

Definition 2.6 [Cun98] Let C be a locally convex algebra. We say that two sub-algebras $A, B \subseteq C$ are **quasi-Morita equivalent** if there exists E and F two closed linear subspaces of C such that the multiplication $m : C \otimes_\pi C \to C$ descends to surjections

$$m_A : E \otimes_\pi F \longrightarrow\!\!\!\!\!\rightarrow A \quad \text{and} \quad m_B : F \otimes_\pi E \longrightarrow\!\!\!\!\!\rightarrow B$$

with bounded linear sections $s_A : A \to E \otimes_\pi F$ and $s_B : B \to F \otimes_\pi E$. We write $A \sim_{q\mathcal{M}} B$ for two quasi-Morita equivalent algebras.

Lemma 2.7. *Let A and B be two locally convex algebras. Suppose it exists a locally convex (A, B)-bimodule E, a locally convex (B, A)-bimodule F and two continuous onto morphisms of A-bimodules and B-bimodules*

$$E \otimes_\pi F \longrightarrow\!\!\!\!\!\rightarrow A \quad \text{and} \quad F \otimes_\pi E \longrightarrow\!\!\!\!\!\rightarrow B$$

with bounded linear sections. Then A and B are quasi-Morita equivalent.

Proof. Consider the the matrix vector space $C = \begin{pmatrix} A & E \\ F & B \end{pmatrix}$. The assumptions of the statement make C be a locally convex matrix-type algebra containing A and B as sub-algebras and E and F as closed linear subspaces. The multiplication on C descends to the bimodule epimorphisms $E \otimes_\pi F \longrightarrow\!\!\!\!\!\rightarrow A$ and $F \otimes_\pi E \longrightarrow\!\!\!\!\!\rightarrow B$ with bounded linear sections. The assertion follows. □

Proposition 2.8. *If A and B are two quasi-Morita equivalent locally convex algebras, then there exists a bounded chain homotopy equivalence*

$$\widehat{CC}(A) \xrightarrow{\sim} \widehat{CC}(B).$$

Proof. Suppose A and B are quasi-Morita equivalent. The continuous morphisms of bimodules $m_A : E \otimes_\pi F \longrightarrow A$ and $m_B : F \otimes_\pi E \longrightarrow B$ are onto and equipped with bounded linear sections. Then the corollary 2.5 applies to the locally convex vector spaces $W = E \otimes_\pi F$ over A and $W = F \otimes_\pi E$ over B. We obtain two chain-homotopy equivalences

$$X\left(T(\widehat{E \otimes_\pi F})\right) \xrightarrow{\sim} \widehat{CC}(A) \quad \text{and} \quad X\left(T(\widehat{F \otimes_\pi E})\right) \xrightarrow{\sim} \widehat{CC}(B)$$

where we considered the adic-completions with respect to the ideals $\ker(m_A)$ and $\ker(m_B)$. According to [CQ95a] and [Cun98], the linear map

$$\begin{array}{ccc} X(T(E \otimes_\pi F)) & \xrightarrow{\Theta} & X(T(F \otimes_\pi E)) \\ (e_0 \otimes f_0) \otimes (e_1 \otimes f_1) \otimes \cdots \otimes (e_n \otimes f_n) & \longmapsto & (f_n \otimes e_0) \otimes (f_0 \otimes e_1) \otimes \cdots \otimes (f_{n-1} \otimes e_n) \\ (e_0 \otimes f_0) \otimes \cdots \otimes (e_n \otimes f_n) d(e_{n+1} \otimes f_{n+1}) & \longmapsto & (f_{n+1} \otimes e_0) \otimes \cdots \otimes (f_{n+1} \otimes e_n) d(f_n \otimes e_{n+1}) \end{array} .$$

defines a continuous isomorphism for the adic and locally convex topologies. Since these X-complexes are dense in their respective adic completions, the assertion follows. □

Definition 2.9 Let G be a real Lie group. We say that two locally convex algebras $A, B \subseteq C$ are **G-equivariantly quasi-Morita equivalent** when G acts smoothly on the algebra C preserving the sub-algebras A, B and the sub-spaces E, F. We write $A \sim_{G-q\mathcal{M}} B$ for two G-equivariant quasi-Morita equivalent algebras.

The notion of G-equivariant quasi-Morita equivalence of algebras enables us to compare not only their respective periodic cyclic complexes as in 2.8, but also the periodic cyclic complexes of their crossed product algebras. Initially established for the action of a discrete group by J. Cuntz [Cun98], the proof extends to the following smooth framework.

Theorem 2.10. *Let G be a real Lie group. If A and B are G-equivariantly quasi-Morita equivalent locally convex algebras, then their smooth crossed products by G are quasi-Morita equivalent:*

$$A \sim_{G-q\mathcal{M}} B \;\Rightarrow\; A \rtimes G \sim_{q\mathcal{M}} B \rtimes G.$$

Proof. We take $E, F \subseteq C$ the G-invariant closed linear subspaces realizing the A and B as G-equivariantly quasi-Morita equivalent algebras. Write δ_e the atomic probability measure centered at the origin of the group G. It fits into the extension

$$0 \longrightarrow \mathcal{C}_c^\infty(G) \longrightarrow M(G) \longrightarrow \mathbb{R} \cdot \delta_e \longrightarrow 0$$

with bounded linear section, where $M(G)$ is the algebra obtained from the convolution algebra $\mathcal{C}_c^\infty(G)$ by adjoining a unit. It can be viewed as the convolution algebra of measures generated by the fdg where f is a compactly supported smooth function and dg a Haar measure, but also by the atomic measure giving to $e \in G$ the weight 1. Taking the tensor product with C preserve the exactness because the first was endowed with a bounded linear section, namely the following is exact

$$0 \longrightarrow C \rtimes G \longrightarrow C' \longrightarrow C \cdot \delta_e \longrightarrow 0$$

where $C' = M(G) \otimes_\pi C$. Since A and B are supposed G-stable the algebras $A \rtimes G$ and $B \rtimes G$ are sub-algebras of $C \rtimes G \subseteq C'$. Take $E' = \mathcal{C}_c^\infty(G, E)$ and $F' = F \cdot \delta_e$

two closed linear subspaces of C'. Then one can check that the multiplication on C' descends to epimorphisms $E' \otimes_\pi F' \longrightarrow\!\!\!\!\!\rightarrow A \rtimes G$ and $F' \otimes_\pi E' \longrightarrow\!\!\!\!\!\rightarrow B \rtimes G$ with sections

$$\begin{array}{rccc} s'_{A\rtimes G}: & A \rtimes G & \longrightarrow & E' \otimes_\pi F' \\ & f & \longmapsto & [g \mapsto (id \otimes \pi(g^{-1}))(s_A(f(g))) \cdot \delta_e] \end{array}$$

$$\begin{array}{rccc} s'_{B\rtimes G}: & B \rtimes G & \longrightarrow & F' \otimes_\pi E' \\ & f & \longmapsto & (\delta_e \otimes id)(s_B(f)) \end{array}$$

Which shows that $A \rtimes G, B \rtimes G \subseteq C'$ are quasi-Morita equivalent. □

Corollary 2.11. *Let G be a real Lie group. If A and B are two G-equivariantly quasi-Morita equivalent locally convex algebras, then there exists a bounded chain homotopy equivalence between the periodic cyclic complexes of their respective smooth crossed product algebras by G*

$$\widehat{CC}(A \rtimes G) \xrightarrow{\sim} \widehat{CC}(B \rtimes G).$$

Proof. It is a direct consequence of the proposition 2.8 and the previous theorem. □

2.3 Green's Imprimitivity theorem

Let G be an almost-conneted real Lie group and fix a left-invariant Haar measure on it. Let also A be a unital locally convex algebra endowed with a smooth action of G. Any closed subgroup $H < G$ acts on $\mathcal{C}_c^\infty(G, A)$ via the action

$$(h \cdot f)(g) := f(gh). \tag{6}$$

We built the crossed product $\mathcal{C}_c^\infty(G, A) \rtimes H$ with respect to this action.

Proposition 2.12. *For any unital locally convex algebra A endowed with a smooth action of G and any closed subgroup $H < G$, the following algebras are G-equivariantly quasi-Morita equivalent*

$$\mathcal{C}_c^\infty(G/H, A) \ \sim_{G-q\mathcal{M}} \ \mathcal{C}_c^\infty(G, A) \rtimes H$$

for the respective G-actions on $\mathcal{A} := \mathcal{C}_c^\infty(G/H, A)$ and $\mathcal{B} := \mathcal{C}_c^\infty(G, A) \rtimes H$ given by

$$(s \cdot a)(gH) = s \cdot a(s^{-1}gH) \quad \text{and} \quad (s \cdot b)(g, h) = s \cdot b(s^{-1}g, h). \tag{7}$$

Proof. Consider the locally convex vector G-spaces $E = F = \mathcal{C}_c^\infty(G, A)$ for the G-actions given by

$$(s \cdot e)(g) = s \cdot e(s^{-1}g) \ \text{ and } \ (s \cdot f)(g) = s \cdot f(s^{-1}g).$$

They are respectively equipped with a $(\mathcal{A}, \mathcal{B})$-bimodule and $(\mathcal{B}, \mathcal{A})$-bimodule structures given by

$$(a \cdot e \cdot b)(g) := \int_H a(gH)e(gh)b(gh, h^{-1})dh \ \text{ and } \ (b \cdot f \cdot a)(g) := \int_H b(g, h)f(gh)a(gH)dh.$$

Indeed, it is clear that the structures of left $\mathcal{A}$-module on E and right $\mathcal{A}$-module on F commute with the pointwise product on $\mathcal{A}$. We now verify that the structures

of right $\mathcal{B}$-module on E and left $\mathcal{B}$-module on F commute with the convolution product on $\mathcal{B}$

$$\begin{aligned}
(e \cdot (b \star b'))(g) &= \int_H e(gh)(b \star b')(gh, h^{-1})dh \\
&= \int_H \int_H e(gh)b(gh, s)b'(ghs, s^{-1}h^{-1})dsdh \\
(hs \to s) &= \int_H \int_H e(gh)b(gh, h^{-1}s)b'(gs, s^{-1})dsdh \\
(h^{-1}s \to h^{-1}) &= \int_H \int_H e(gsh)b(gsh, h^{-1})b'(gs, s^{-1})dsdh \\
&= \int_H (e \cdot b)(gs)b'(gs, s^{-1})ds \\
&= ((e \cdot b) \cdot b')(g)
\end{aligned}$$

$$\begin{aligned}
((b \star b') \cdot f)(g) &= \int_H (b \star b')(g, h)f(gh)dh \\
&= \int_H \int_H b(g, s)b'(gs, s^{-1}h)f(gh)dsdh \\
(s^{-1}h \to h) &= \int_H \int_H b(g, s)b'(gs, h)f(gsh)dsdh \\
&= \int_H b(g, s)(b' \cdot f)(gs)ds \\
&= (b \cdot (b' \cdot f))(g)
\end{aligned}$$

One can also show that these bimodule structures are also compatible with the actions of G by direct computation. Define the two continuous morphisms of A-bimodules an B-bimodules respectively

$$\begin{array}{rccc}
m_{\mathcal{A}} : & E \otimes_\pi F & \longrightarrow & \mathcal{A} \\
& (e, f) & \longmapsto & [gH \mapsto \int_H e(gh)f(gh)dh]
\end{array}$$

$$\begin{array}{rccc}
m_{\mathcal{B}} : & F \otimes_\pi E & \longrightarrow & \mathcal{B} \\
& (f, e) & \longmapsto & [(g, h) \mapsto f(g)e(gh)]
\end{array}.$$

We shall find bounded sections of these morphisms, to apply the lemma 2.7 and get expected result. Take a locally finite covering $\{U_i\}_i$ of G and consider a set $\{\rho_i\}_i$ of compactly supported smooth functions on G subordinate to the covering such that

$$\sum_i \left(\int_H \rho_i(gh)dh \right) = 1.$$

Their existence is provided by the fact that G is locally compact and H is closed. See [Bru61] for their construction. Also, consider a set $\{\epsilon_i\}_i$ of compactly supported smooth functions on G such that $\epsilon_i \equiv 1$ on the support of ρ_i. Now, define the linear map

$$\begin{array}{rccc}
s_{\mathcal{A}} : & \mathcal{A} & \longrightarrow & E \otimes_\pi F \\
& a & \longmapsto & [(g, h) \mapsto \rho_i(g)a(gH) \otimes \epsilon_i(h) \cdot 1_A]
\end{array}.$$

It is obviously continuous and its image is compactly-supported because both ρ_i and ϵ_i are for all i. It defines a bounded section of $m_{\mathcal{A}}$ as shows the following

computation

$$(m_{\mathcal{A}} \circ s_{\mathcal{A}})(a)(gH) = \sum_i \left(\int_H \rho_i(gh) a(gH) \epsilon_i(gh) dh \right)$$
$$= a(gH) \sum_i \left(\int_H \rho_i(gh) \epsilon_i(gh) dh \right) = a(gH)$$

by definition of the ρ_i and ϵ_1. Finally, the linear map

$$\begin{array}{rccc} s_{\mathcal{B}} : & \mathcal{B} & \longrightarrow & F \otimes_\pi E \\ & b & \longmapsto & [(g,h) \mapsto b(g, g^{-1}h)] \end{array} .$$

is a bounded section of $m_{\mathcal{B}}$ since $(m_{\mathcal{B}} \circ s_{\mathcal{B}})(b)(g,h) = s_B(b)(g,gh) = b(g,h)$. We showed that $s_{\mathcal{A}}$ is a bounded section of $m_{\mathcal{A}}$ and $s_{\mathcal{B}}$ is a bounded section of $m_{\mathcal{B}}$, which gives the expected result since everything is compatible with the G-actions. □

Theorem 2.13 (Weak version of Green's imprimity theorem)**.** *Let G be a real Lie group and $H < G$ a closed subgroup of it. For every unital locally convex algebra A endowed with a smooth action of G, there exists an isomorphism in periodic cyclic homology*

$$HP_\bullet(A \rtimes H) \xrightarrow{\sim} HP_\bullet(\mathcal{C}_c^\infty(G/H, A) \rtimes G).$$

Proof. The previous proposition applied to G and the subgroup H yields the following equivariant quasi-Morita equivalences

$$A \ \sim_{H-q\mathcal{M}} \ \mathcal{C}_c^\infty(G,A) \rtimes G \ \text{ and } \ \mathcal{C}_c^\infty(G/H,A) \sim_{G-q\mathcal{M}} \ \mathcal{C}_c^\infty(G,A) \rtimes H.$$

The first being a G-equivariant quasi-Morita equivalence it is in particular H-equivariant. Due to the corollary 2.11, it induces some chain-homotopy equivalences

$$\widehat{CC}(A \rtimes H) \xrightarrow{\sim} \widehat{CC}\Big((\mathcal{C}_c^\infty(G,A) \rtimes G) \rtimes H \Big) \ \text{ and } \ \widehat{CC}\big(\mathcal{C}_c^\infty(G/H,A) \rtimes G\big) \xrightarrow{\sim} \widehat{CC}\Big((\mathcal{C}_c^\infty(G,A) \rtimes H) \rtimes G \Big).$$

We can rewrite these *double* crossed products as crossed product

$$(\mathcal{C}_c^\infty(G,A) \rtimes G) \rtimes H \simeq \mathcal{C}_c^\infty(G,A) \rtimes_\alpha (G \times H) \ \text{ and } \ (\mathcal{C}_c^\infty(G,A) \rtimes H) \rtimes G \simeq \mathcal{C}_c^\infty(G,A) \rtimes_\beta (G \times H)$$

associated to $G \times H$ for the actions α and β given, for $(s,h) \in G \times H$, by the following formulas

$$\alpha_{(s,h)} f(g) = h \cdot f(h^{-1} g s) \ \text{ and } \ \beta_{(s,h)} f(g) := s \cdot f(s^{-1} g h)$$

according to the (6) and (7). The proof follows from an identification of these actions under the linear map $\varphi : \mathcal{C}_c^\infty(G,A) \longrightarrow \mathcal{C}_c^\infty(G,A)$ defined by the formula $\varphi(f)(g) = g \cdot f(g^{-1})$. Indeed, it is a continuous isomorphism of locally convex vector spaces which commutes with the pointwise multiplication on $\mathcal{C}_c^\infty(G,A)$ and transports the $G \times H$-action α to the $G \times H$-action β

$$\varphi(\alpha_{(s,h)} f)(g) = g \cdot \alpha_{(s,h)} f(g^{-1}) = gh \cdot f(h^{-1} g^{-1} s) = s \cdot \varphi(f)(s^{-1} g h) = \beta_{(s,h)} \varphi(f)(g).$$

The assertion follows. □

3 Dirac-dual Dirac method in KK-theory

Let G be an almost-connected real Lie group, K a maximal compact subgroup of it and A a unital locally convex algebra endowed with a smooth action of G. The aim of this section is to introduce the main results required to show the isomorphism (4).

Together with the theorem 2.13 we shall show in the next section that there exists an isomorphism

$$HP_\bullet(\mathcal{A} \rtimes K) \xrightarrow{\sim} HP_{\bullet+\dim(G/K)}(\mathcal{A} \rtimes G) \tag{8}$$

after stabilization $\mathcal{A} = A \otimes_\pi \mathcal{D}$ with a certain Schatten ideal $\mathcal{D}$.

Lemma 3.1. *If the morphism (8) above is an isomorphism for any pair (G, K) with G/K even-dimensional, then it is also an isomorphism for the pairs (G, K) such that G/K is odd-dimensional.*

Proof. Take a pair (G, K) such that G/K is odd dimensional. Any maximal compact subgroup of $G' := G \times \mathbb{R}$ is conjugated to K and the quotient space G'/K is homeomorphic to $G/K \times \mathbb{R}$ of even dimension. By assumption, the map (8) above is then an isomorphism for the pair (G', K). The assertion follows from a Connes-Thom isomorphism $HP_\bullet(\mathcal{A} \rtimes G') \xrightarrow{\sim} HP_{\bullet+1}(\mathcal{A} \rtimes G)$ in periodic cyclic homology, see [Nis93, §4.10] and [ENN88]. □

Assumption: In the whole section, the homogeneous space G/K **will be supposed even-dimensional** as a consequence of the previous lemma. In our case of interest, the isomorphism (4) becomes, after stabilization

$$HP_\bullet(\mathcal{C}_c^\infty(G/K, \mathcal{A}) \rtimes G) \xrightarrow{\sim} HP_\bullet(\mathcal{A} \rtimes G). \tag{9}$$

The idea to obtain this isomorphism (in the even-dimensional case) is to find a G-equivariant morphism of locally convex algebras whose crossed product by G is a quasi-isomorphism in periodic cyclic homology. Indeed, the theorem 1.1 of V. Nistor [Nis93] establishes a condition to get a quasi-isomorphism after passing to crossed product.

Corollary 3.2 (Nistor)**.** *Let G be an almost connected-Lie group and K a maximal compact subgroup of it. Suppose $\varphi : A \to B$ is a continuous morphism of locally convex algebras endowed with smooth actions of G. Then*

$$HP(\varphi \rtimes K) \text{ is invertible } \Rightarrow \ HP(\varphi \rtimes G) \text{ is an isomorphism.}$$

Proof. The map φ extends into morphisms $\varphi \rtimes G : A \rtimes G \to B \rtimes G$ and $\varphi \rtimes K : A \rtimes K \to B \rtimes K$ as it is G-equivariant and in particular K-equivariant. The theorem 1.1 of Nistor tells us that for any maximal ideal $\mathfrak{m}$ of the smooth class functions algebra $\mathcal{C}^\infty(G)^{\text{inv}}$ we have the following isomorphism, natural in A

$$HP_\bullet(A \rtimes G)_\mathfrak{m} \simeq HP_{\bullet+\dim(G/K)}(A \rtimes K)_\mathfrak{m}.$$

Using this theorem for the G-algebras A and B builds the following commutative diagram with vertical isomorphisms

$$\begin{array}{ccc} HP_\bullet(A \rtimes G)_\mathfrak{m} & \xrightarrow{HP(\varphi \rtimes G)} & HP_\bullet(B \rtimes G)_\mathfrak{m} \\ \sim\downarrow & & \downarrow\sim \\ HP_{\bullet+\dim(G/K)}(A \rtimes K)_\mathfrak{m} & \xrightarrow{HP(\varphi \rtimes K)} & HP_{\bullet+\dim(G/K)}(B \rtimes K)_\mathfrak{m} \end{array}$$

Indeed, as $HP(\phi \rtimes K)$ is supposed to be invertible, the bottom arrow becomes an isomorphism, and so is the the top arrow. We showed that $\varphi \rtimes G$ is globally defined and a quasi-isomorphism on every conjuacy class of G, and thus defines a global quasi-isomorphism. □

Strategy 3.3. *Our strategy then relies on the existence of:*

- *a G-equivariant morphism α of locally convex algebras inducing the linear map*
$$HP(\alpha \rtimes G) : HP_\bullet(\mathcal{C}_c^\infty(G/K, A) \rtimes G) \longrightarrow HP_\bullet(A \rtimes G);$$
- *a K-equivariant morphism β of locally convex algebras such that $HP(\beta \rtimes K)$ is an inverse of $HP(\alpha \rtimes K)$.*

In that case, $HP(\alpha \rtimes G)$ would realize the expected isomorphism (9).

The morphisms α and β, which can be thought as K-equivariantly dual to each other, *implement Dirac and dual Dirac elements in periodic cyclic homology.* In this section we propose an introduction to the main results of the Dirac-dual Dirac construction in bivariant K-theory.

3.1 Dirac and dual Dirac elements

This section introduces the construction of the Dirac and dual Dirac as bivariant K-theory classes and highlight their behavior regarding to the Kasparov product.

Let q be an integer. The **Clifford algebra** of $\mathbb{R}^q$ is the *freest* unital associative algebra generated by $\mathbb{R}^q$ with product subject to the condition $x^2 = \|x\|^2 \cdot 1$, for all $x \in \mathbb{R}^q$. It is defined as

$$\mathrm{Cl}(q) := T\mathbb{R}^q / \langle x \otimes x - \|x\|^2 \cdot 1,\ x \in \mathbb{R}^q \rangle.$$

If $e_1, \cdots, e_q$ is an orthogonal basis of $\mathbb{R}^q$, a basis of the Clifford algebra as a real vector space is given by $\{e_{i_1} \cdots e_{i_k} \mid 1 \leq i_1 < \cdots < i_k \leq q \text{ and } 0 \leq k \leq q\}$ and $\dim_{\mathbb{R}}(\mathrm{Cl}(q)) = 2^q$. The Clifford algebra decomposes into some even-odd parts

$$\mathrm{Cl}(q) = \mathrm{Cl}(q)_+ \oplus \mathrm{Cl}(q)_- \tag{10}$$

generated by respectively even and odd elements of the basis. Let us use the notation $\mathrm{Cl}_{\mathbb{C}}(q) = \mathrm{Cl}(q) \otimes_{\mathbb{R}} \mathbb{C}$ for its complexification.

Recall that the homogeneous space G/K is a Euclidian space whose dimension (even by assumption of the section §3) will be denoted $q = \dim(G/K)$. Let us call **Clifford module** and **smooth Clifford module** of G/K the following algebras

$$\mathcal{C}_\tau(G/K) = \mathcal{C}_0(G/K, \mathrm{Cl}_{\mathbb{C}}(q)) \quad \text{and} \quad \mathcal{C}_\tau^\infty(G/K) = \mathcal{C}_c^\infty(G/K, \mathrm{Cl}_{\mathbb{C}}(q)).$$

These are algebras for the pointwise product $(s \cdot s')(x) = s(x) \cdot s'(x)$ where $\cdot$ denotes the Clifford multiplication.

The Riemannian connection ∇ acts as a derivation on the smooth Clifford module $\mathcal{C}_\tau^\infty(G/K)$. We endow this algebra with Fréchet topology coming from the family of $\mathcal{C}^k$-sup norms

$$\|f\|_k := \|\nabla^k(f)\| \tag{11}$$

with $k \in \mathbb{N}$ controlling the growth of the derivatives. The Clifford module, as for him, is the $C^\star$-completion of the smooth Clifford module. Both of these algebra are endowed with the smooth action of G given by left translation.

Theorem 3.4. *For every complete unital locally convex algebra A endowed with a smooth action of G there exists a continuous chain-homotopy equivalence*

$$\widehat{CC}_\bullet((C_c^\infty(G/K)\widehat{\otimes}_\pi A) \rtimes G) \simeq \widehat{CC}_\bullet((C_\tau^\infty(G/K)\widehat{\otimes}_\pi A) \rtimes G).$$

Proof. We shall show that $\mathcal{C}_c^\infty(G/K)\widehat{\otimes}_\pi A$ and $\mathcal{C}_\tau^\infty(G/K)\widehat{\otimes}_\pi A$ are G-equivariantly quasi-Morita equivalent in order to use corollary 2.11. Since G/K is even-dimension by assumption 3, there exists a G-equivariant vector bundle $\mathcal{S}$ over G/K such that the Clifford multiplication provides a G-equivariant continuous isomorphism $\mathrm{Cl}_{\mathbb{C}}(TG/K) \xrightarrow{\sim} \mathrm{End}(\mathcal{S})$, and induces is an isomorphism of G-algebras

$$C_\tau^\infty(G/K) \xrightarrow{\sim} \Gamma_c^\infty(G/K\, \mathrm{End}(\mathcal{S}))$$

continuous for the locally Fréchet topologies. Now, consider the locally convex spaces $E := \Gamma_c^\infty(G/K, \mathcal{S})$ and $F := \Gamma_c^\infty(G/K, \mathcal{S}^*)$. It is easy to check that the multiplication makes E a $(\mathcal{C}_\tau^\infty(G/K), \mathcal{C}_c^\infty(G/K))$-bimodule and F a $(\mathcal{C}_c^\infty(G/K), \mathcal{C}_\tau^\infty(G/K))$-bimodule. The following map are continuous onto morphism of bimodules and admit continuous linear sections:

$$\begin{aligned} E\widehat{\otimes}_\pi F &\longrightarrow \mathcal{C}_\tau^\infty(G/K), & s \otimes \lambda &\longmapsto [x \mapsto (v \mapsto s(x)\lambda(x)(v))], \\ F\widehat{\otimes}_\pi E &\longrightarrow \mathcal{C}_c^\infty(G/K), & \lambda \otimes s &\longmapsto [x \mapsto \lambda(x)(s(x))]. \end{aligned}$$

Since the actions of G on $\mathcal{S}$ and $\mathcal{S}^*$ preserve the module structures and the two pairings, the algebras $\mathcal{C}_c^\infty(G/K)$ and $\mathcal{C}_\tau^\infty(G/K)$ are G-equivariantly quasi-Morita equivalent. Finally, since A a unital locally convex G-algebra, the tensor product algebras $\mathcal{C}_c^\infty(G/K)\widehat{\otimes}_\pi A$ and $\mathcal{C}_\tau^\infty(G/K)\widehat{\otimes}_\pi A$ remain G-equivariant quasi-Morita equivalent. □

In the two next paragraphs, we will define the Dirac and dual Dirac elements as classes in equivariant KK-groups associated to $\mathcal{C}_\tau(G/K)$.

3.1.1 The Dirac element

This paragraph provides a definition of the Dirac element as a G-equivariant Fredholm module over $\mathcal{C}_\tau(G/K)$, namely a class $\alpha \in KK_G(\mathcal{C}_\tau(G/K), \mathbb{C})$. We recall here the analytic model for equivariant K-homology from the references [Val02] and [HR01].

Definition 3.5 Let G be a locally compact group and A a $C^\star$-algebra endowed with a continuous action of G. A **G-equivariant Fredholm module** over A is a triplet (ρ, F, U) where:

- $\rho : A \to \mathcal{L}(\mathcal{H})$ is a bounded $\star$-representation into a Hilbert space $\mathcal{H}$;
- F a self-adjoint bounded operator on $\mathcal{H}$;
- $U : G \to U(\mathcal{H})$ is a unitary representation;

which verify for any $g \in G$ and $a \in A$ that

$$\rho(a)(F^2-1) \in \mathcal{K}(\mathcal{H}), \quad \rho(a)F - F\rho(a) \in \mathcal{K}(\mathcal{H}), \quad U_g F = F U_g \text{ and } U_g \rho(a) U_g^{-1} = \rho(g \cdot a). \tag{12}$$

A G-equivariant Fredholm module (ρ, F, U) is **even** if the underlying Hilbert space decomposes as $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$, ρ and U are degree preserving and F is odd. Otherwise, the Fredholm module is said to be **odd**. Two G-equivariant Fredholm modules (ρ, F, U) and (ρ', F', U') are **operator homotopic** if $\rho = \rho'$, $U = U'$ and if there exists a norm continuous path $(F_t)_{t\in[0,1]}$ connecting F to F' such that for each $t \in [0,1]$ and $g \in G$, $U_g F_t = F_t U_g$.

Definition 3.6 We define the group $KK_G(A, \mathbb{C})$ to be the abelian group of operator homotopy classes of even G-equivariant Fredholm modules over A.

Let us fix $\mathcal{H} = L^2(\Omega^\star(G/K))$ to be the L^2-completion of the compactly supported differential forms $\Omega_c^\star(G/K)$ over the homogeneous space G/K. Under the identification of the Clifford algebra $\mathrm{Cl}(q)$ with the exterior algebra $\bigwedge \mathbb{R}^q$ it is clear that $\mathcal{H}$ is a module over the $C^\star$-algebra $\mathcal{C}_\tau(G/K)$. Namely, the Clifford multiplication induces a bounded $\star$-representation

$$\begin{array}{rccc} \rho: & \mathcal{C}_\tau(G/K) & \longrightarrow & \mathcal{L}(\mathcal{H}) \\ & s & \longmapsto & f \mapsto [x \mapsto s(x) \cdot f(x)] \end{array}.$$

Consider also the De-Rham Laplace operator $D := d + d^\star : \mathcal{H} \longrightarrow \mathcal{H}$. Its *bounded version* is a self-adjoint bounded operator on $\mathcal{H}$

$$F := D(1+D^2)^{-1/2} \in \mathcal{L}(\mathcal{H}).$$

Finally, the real Lie group G acts by left-translation on the homogeneous space G/K and preserves the metric. It induces a unitary representation of the group on $\mathcal{H}$ given by:

$$\begin{array}{rccc} U: & G & \longrightarrow & U(\mathcal{H}) \\ & g & \longmapsto & f \mapsto [x \mapsto f(g \cdot x)] \end{array}.$$

Theorem 3.7 (Kasparov, [Kas88])**.** *Let G be an almost-connected real Lie group. The triplet (ρ, F, U) described above is an even G-equivariant Fredholm module over the $C^\star$-algebra $\mathcal{C}_\tau(G/K)$, hence represents a class $\alpha := [(\rho, F, U)] \in KK_G(\mathcal{C}_\tau(G/K), \mathbb{C})$ called **Dirac element** of G.*

3.1.2 The dual Dirac element

Now we propose a definition for the dual Dirac element as a class in the K-equivariant K-theory of $\mathcal{C}_\tau(G/K)$, which is a class $\beta \in KK_K(\mathbb{C}, \mathcal{C}_\tau(G/K))$. We recall a model for equivariant K-theory.

Let A be a $C^\star$-algebra. We say that two matrices $p \in M_n(A)$ and $q \in M_m(A)$ are **stable-equivalent** if there exists an integer $r \geq \max(m,n)$ and a matrix $u \in GL_r(A)$ such that:

$$u \begin{pmatrix} p & 0 \\ 0 & 0_{r-n} \end{pmatrix} u^{-1} = \begin{pmatrix} q & 0 \\ 0 & 0_{r-m} \end{pmatrix}.$$

If K is a compact group acting continuously on A, the set of stable-equivalent classes of K-equivariant idempotent matrices over A (for the entrywise action) is a monoid for the block-diagonal direct sum.

Definition 3.8 We define the group $KK_K(\mathbb{C}, A)$ to be the abelian Grothendieck group of stable-equivalent classes of K-equivariant idempotent matrices over A.

Consider a smooth function $\chi : \mathbb{R}^q \to \mathbb{R}$ invariant under rotations which is identically 1 on the ball $B(0, 1/2)$ and identically zero out of $B(0, 1)$. We define two smooth maps

$$e_0, e_1 : \mathbb{R}^q \longrightarrow M_2(\mathrm{Cl}_{\mathbb{C}}(q))$$

which associate to any point $x \in \mathbb{R}^q$ the idempotent matrices

$$e_0(x) = \frac{1}{\chi(x)^2 + \|x\|^2} \begin{pmatrix} \chi(x)^2 & \chi(x) \cdot x \\ \chi(x) \cdot x & \|x\|^2 \end{pmatrix} \quad \text{and} \quad e_1(x) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Lemma and Definition 3.9. *The smooth functions above belong to the matrix space of the unitalization of $\mathcal{C}^\infty_\tau(G/K)$, namely $e_0, e_1 \in M_2(\mathcal{C}^\infty_\tau(G/K)^+)$, and their difference induces a class $[e_0] - [e_1] \in KK_K(\mathbb{C}, \mathcal{C}^\infty_\tau(G/K))$. We define the **dual Dirac element** of G to be the class $\beta \in KK_K(\mathbb{C}, \mathcal{C}_\tau(G/K))$ obtained after $C^\star$-completion of smooth Clifford module.*

Proof. For any $x \in \mathbb{R}^q$, $e_0(x)$ and $e_1(x)$ are idempotent matrices over $\mathrm{Cl}_{\mathbb{C}}(q)$ so e_0 and e_1 are idempotent matrices over the algebra of continuous functions $\mathcal{C}^0(\mathbb{R}^q, \mathrm{Cl}_{\mathbb{C}}(q))$. When x doesn't belong to unit-ball, $e_0(x) = e_1(x)$ by definition of χ, which makes their difference $e_0 - e_1$ a matrix with compactly supported coefficients, *i.e.* an element of $M_2(\mathcal{C}^\infty_c(\mathbb{R}^q, \mathrm{Cl}_{\mathbb{C}}(q)))$. One can always choose an homeomorphism from the homogeneous space G/K to the Euclidian space $\mathbb{R}^q$ which transforms the translation K-action on G/K into the rotation-action on $\mathbb{R}^q$. Under this homeomorphism both e_0 and e_1 belong to $M_2(\mathcal{C}^\infty_\tau(G/K)^+)$ and commute with the action of K. In particular $e_0 - e_1 \in M_2(\mathcal{C}^\infty_\tau(G/K))$ is a difference of K-equivariant idempotent matrices, which yields the proof. □

Theorem 3.10 (Kasparov [Kas81])**.** *The Kasparov product of the dual Dirac element β with the Dirac element α is K-equivariantly trivial in both direction*

$$\mathrm{Res}^G_K(\alpha) \times \beta = 1 \in KK_K(\mathcal{C}_\tau(G/K), \mathcal{C}_\tau(G/K)) \quad \textit{and} \quad \beta \times \mathrm{Res}^G_K(\alpha) = 1 \in KK_K(\mathbb{C}, \mathbb{C}).$$

Remark We stated here the Kasparov's theorem in a weaker form because we only need the dual Dirac element to be K-equivariant in our setup. Their products remain trivial in both direction after restriction to $K < G$.

The Dirac and dual Dirac elements being thought as K-equivariant dual to each others, they will produce the K-equivariant dual morphisms of locally convex algebras α and β we want to obtain in our strategy 3.3. The next section proposes a way to realize these bivariant K-theory classes as $\star$-homomorphisms.

3.2 Cuntz' approach to KK-theory

We would like to get algebraic description of the Dirac and dual Dirac elements in order to adapt their construction to periodic cyclic homology. This algebraic description is provided by the Cuntz-picture which identifies bivariant theory groups

$KK(A,B)$ as homotopy classes of $\star$-homomorphisms from the Cuntz' algebra qA to $B \otimes_{C^\star} \mathcal{K}$, where $\mathcal{K}$ denote the $C^\star$-algebra of compact operators on a infinite dimensional Hilbert space. In this setting, the Kasparov product corresponds to the composition of the underlying homomorphisms.

Cuntz' algebra

Let A and B be two complex algebras. The **algebraic free product** $A \star B$ is the algebra generated by words with letters alternating between A and B, modulo the relations internal to each algebra:

$$A \star B := A \oplus B \oplus (A \otimes B) \oplus (B \otimes A) \oplus (A \otimes B \otimes A) \oplus (B \otimes A \otimes B) \oplus \cdots$$

It is naturally equipped with two canonical inclusions

$$\iota_0 : A \to A \star B \quad \text{and} \quad \iota_1 : B \to A \star B. \tag{13}$$

When $A = B$, we define the **free product** QA to be the universal ring generated by the two copies $\iota_0(A)$ and $\iota_1(A)$. Namely, it is generated by the reduced words

$$\iota_0(a_0)\iota_1(a_1)\iota_0(a_2)\cdots\iota_{n \bmod 2}(a_n) \in QA, \quad a_i \in A. \tag{14}$$

The free product verifies the semi-universal property that any pair of homomorphisms $\phi, \psi : A \to B$ induces a homomorphism $\phi \star \psi : QA \to B$ such that $(\phi \star \psi) \circ \iota_0 = \phi$ and $(\phi \star \psi) \circ \iota_1 = \psi$.

Definition 3.11 The **Cuntz's algebra** qA associated to an algebra A is the kernel of the fold map $id \star id : QA \to A$ arising from the pair of identity morphisms

$$qA := \ker(id \star id : QA \to A).$$

It fits into the following doubly-split extension of algebras

$$0 \longrightarrow qA \hookrightarrow QA \underset{\iota_0,\, \iota_1}{\overset{id\star id}{\rightleftarrows}} A \longrightarrow 0 \ .$$

If the algebra A is endowed with a $C^\star$-structure, then its free product QA is also a $C^\star$-algebra for the largest possible $C^\star$-norm, namely:

$$\|x\|_{QA} := \{\|\pi(x)\| \mid \pi : QA \to \mathcal{L}(H) \text{ is a } \star\text{-representation into a Hilbert space } H\}.$$

The Cuntz's algebra qA being a subspace of it, it inherits the induced topology. It verifies the following fundamental property [Cun87].

Theorem 3.12 (Cuntz)**.** *For any separable $C^\star$-algebra A, there exists an homotopy equivalence $q(qA) \xrightarrow{\sim} qA$ after stabilization by square matrices.*

Cuntz' identification

The Cuntz' approach was motivated by the search of an algebraic description of the bivariant K-theory. Indeed, K-theory and K-homology groups should both encode differences of $\star$-homomorphisms, and the kernel qA of the fold map is precisely the universal $C^\star$-algebra in which such differences become genuine algebraic elements. It is the universal receptacle of difference of $\star$-homomorphisms out of A. This identifies qA as the natural object representing bivariant K-theory.

Theorem 3.13 (Cuntz, [Cun87]). *Let G be a locally compact group and (A, B) a pair of separable $C^\star$-algebras endowed with a continuous action of G. Any G-equivariant KK-theory class uniquely determines a continuous G-equivariant $\star$-homomorphism modulo homotopy*

$$KK_G(A, B) \longrightarrow \mathrm{Hom}_G(qA, B \otimes_{C^\star} \mathcal{K})_{/\sim}.$$

If the group is compact, this description is a one-to-one correspondence.

This approach provides a simple description of the equivariant Kasparov product. Given A, B and C three $C^\star$-algebras endowed with a continuous action of a compact group K we can *compose* some continuous K-equivariant maps $f : qA \to B \otimes_{C^\star} \mathcal{K}$ and $g : qB \to C \otimes_{C^\star} \mathcal{K}$ in the following way

$$g \circ qf : qA \xrightarrow[3.12]{\sim} q(qA) \longrightarrow q(B \otimes_{C^\star} \mathcal{K}) \longrightarrow C \otimes_{C^\star} \mathcal{K} \otimes_{C^\star} \mathcal{K} \simeq C \otimes_{C^\star} \mathcal{K}. \quad (15)$$

This composition defines a class in $KK_K(A, C)$ which depends only on the homotopy classes of f and g.

Theorem 3.14 (Cuntz, [Cun87]). *Given two classes $[f] \in KK_K(A, B)$ and $[g] \in KK_K(B, C)$, there exists an homotopy equivalence between their Kasparov product and their composition in the sense of (15). Namely, these two computations define the same class in bivariant K-theory*

$$[f] \times [g] = [g \circ qf] \in KK_K(A, C).$$

Composition of Dirac and dual Dirac elements

Consider a Hilbert space $\mathcal{H}$ endowed with a unitary representation $U : G \to \mathcal{L}(\mathcal{H})$ of a real Lie group G. We define the **G-smooth p^{th}-Schatten ideal** of $\mathcal{H}$ as the ideal:

$$\ell^{p,G}(\mathcal{H}) := \left\{ T \in \mathcal{L}(\mathcal{H}) \;\middle|\; \begin{array}{l} T \in \ell^p(\mathcal{H}), \text{ namely } \mathrm{Tr}((T^\star T)^p) < \infty \\ g \mapsto U(g) T U(g)^{-1} \text{ is infinitely differentiable for all } g \in G \end{array} \right\}. \quad (16)$$

It is the greatest subalgebra of the pth-Schatten ideal $\ell^p(\mathcal{H})$ which is endowed with a smooth action of G.

Lemma 3.15. *(Kasparov [Kas81]) Let G be an almost-connect real Lie group and K a maximal compact subgroup of it. Let $P \in \ell^{p,G}(\mathcal{H})$ be a projection from $\mathcal{H}$ onto a one-dimensional subspace of K-fixed vectors. The trivial class $1 = [\mathbb{1}_K] \in KK_K(\mathbb{C}, \mathbb{C})$ can be represented by the composition of algebra homomorphisms*

$$\begin{array}{rccc} \mathbb{1}_K : q(q\mathbb{C}) & \xrightarrow{(id\star 0)\circ(id\star 0)} & \mathbb{C} & \xrightarrow{\quad\iota\quad} & M_2(\ell^{p,G}(\mathcal{H})) \\ & & 1 & \longmapsto & \begin{pmatrix} P & 0 \\ 0 & 0 \end{pmatrix} \end{array}.$$

Recall that the Dirac element $\alpha \in KK_G(\mathcal{C}_\tau(G/K), \mathbb{C})$ and the dual Dirac element $\beta \in KK_K(\mathbb{C}, \mathcal{C}_\tau(G/K))$ define equivariant KK-classes as established before. The theorem 3.13 shows that they can be represented as a G-equivariant and a K-equivariant $\star$-homomorphisms

$$\alpha : q(\mathcal{C}_\tau(G/K)) \longrightarrow \mathcal{K} \quad \text{and} \quad \beta : q\mathbb{C} \longrightarrow M_2(\mathcal{C}_\tau(G/K)).$$

The image of the dual Dirac element lies in the algebra of 2×2 matrices since it is defined using matrices of this size (3.9).

Corollary 3.16. *The following morphisms of algebras are K-equivariantly homotopy equivalent to each others*

$$\alpha \circ q\beta : q(q\mathbb{C}) \longrightarrow M_2(\mathcal{K}) \quad \textit{and} \quad \mathbb{1}_K : q(q\mathbb{C}) \longrightarrow M_2(\ell^{p,G}(\mathcal{H})) \subseteq M_2(\mathcal{K}).$$

Proof. By the Kasparov theorem 3.10 we know that the Kasparov product $\beta \times \mathrm{Res}_K^G(\alpha)$ is trivial. Then, the theorem 3.14 shows that their composition $\alpha \circ q\beta$ is K-equivariantly homotopy equivalent to a representative of the trivial class in K-equivariant KK-theory. The assertion follows from the lemma abvove. □

Remark In the paper [**Gre**], the author shows that in the case of Euclidian Dirac and dual Dirac elements

$$\alpha^{\mathrm{Euc}} \in KK_G(\mathcal{C}_\tau(\mathbb{R}^q), \mathbb{C}) \quad \text{and} \quad \beta^{\mathrm{Euc}} \in KK_K(\mathbb{C}, \mathcal{C}_\tau(\mathbb{R}^q)), \tag{17}$$

the K-equivariant continuous-homotopy between the Kasparov composition $\alpha^{\mathrm{Euc}} \circ q\beta^{\mathrm{Euc}}$ and the map $\mathbb{1}_K$ can be chosen to be smooth. We will use this stronger statement on 4.11 because periodic cyclic homology is only stable under smooth-homotopies.

4 Dirac-dual Dirac method in periodic cyclic homology

The aim of this section is to show that the Dirac element $\alpha \in KK_G(\mathcal{C}_\tau(G/K), \mathbb{C})$ realizes the expected isomorphism (9) of periodic cyclic homology groups

$$HP((\alpha \otimes id_{\mathcal{A}}) \rtimes G) : HP_\bullet(\mathcal{C}_c^\infty(G/K, \mathcal{A}) \rtimes G) \xrightarrow{\sim} HP_\bullet(\mathcal{A} \rtimes G)$$

after stabilization $\mathcal{A} = A \otimes_\pi \mathcal{D}$ with a certain Schatten ideal $\mathcal{D}$ (see 4.12 for its definition). To obtain this result, we will *descend* the construction of the previous section from bivariant K-theory to periodic cyclic homology by refining the Cuntz identification 3.13 for the Dirac and dual Dirac elements. We shall show that we can associate to them continuous homomorphism of locally convex algebras that are K-equivariantly inverses to each other at the level of periodic cyclic complexes, as expected in the strategy 3.3.

4.1 Rescaled version of the Cuntz algebra

We first adapt the Cuntz' algebra qA defined in the previous section into a rescaling family of algebras $\{q_R A\}_{R>0}$ playing an analogous role for bounded homomorphisms of Fréchet algebras instead of $C^\star$-homomorphisms.

The rescaled free product

Let $(A, \{\|\cdot\|_k\}_{k\geq 0})$ be a Fréchet algebra endowed with a countable family of seminorms. Recall that its algebraic free product $A \star A$ is the universal ring generated by the image of the canonical inclusion ι_0 and ι_1 defined in (13). We endow the algebraic free product with a Fréchet structure as follows.

Fix $R \geq 1$ and an integer $m \in \mathbb{N}$, the (R, m)**-norm** is defined on every reduced words $x := \iota_0(a_0) \cdots \iota_{n \bmod 2}(a_n) \in A \star A$ (14) by the following formula

$$\|x\|_{(R,m)} := \sum_{m_0 + \cdots + m_n \leq m} (n+1) R^n \|a_0\|_{m_0} \cdots \|a_n\|_{m_n}. \tag{18}$$

Definition 4.1 For a fixed $R \geq 1$ we define the **R-free product** of a Fréchet algebra A as the completion $Q_R A$ of the algebraic free product $A \star A$ with respect to the family of semi-norms $\{\|\cdot\|_{(R,m)}\}_{m\geq 0}$.

Suppose that the Fréchet algebra A is endowed with a smooth action of a real Lie group G. For any $R \geq 0$, we extend the action of G on $Q_R A$ coordinatewise, which in on reduced words

$$g \cdot \Big(\iota_0(a_0) \cdots \iota_{n \bmod 2}(a_n)\Big) := \iota_0(g \cdot a_0) \cdots \iota_{n \bmod 2}(g \cdot a_n).$$

This action is not necessarily smooth.

Theorem 4.2. *Suppose A is a Fréchet algebra. The two inclusions on first and second coordinates $j_0, j_1 : A \rightrightarrows A \oplus A$ induce, for all $R \geq 1$, a continuous morphism of Fréchet algebras*

$$i := j_0 \star j_1 : Q_R A \to A \oplus A$$

which is invertible modulo smooth homotopy after stabilization by 2×2-matrices. Moreover if A is endowed with a smooth action of a real Lie group G which extends smoothly on $Q_R A$, then the smooth homotopy is also G-equivariant.

Proof. It is clear that, for every $N \geq 0$ and $a \in A$, one has $\|j_i(a)\|_N \leq \|a\|_N$ for $i = 0, 1$. Then by the previous proposition, the free product algebra homorphism extends for every $R \geq 1$ as a continuous algebra homomorphism

$$i := j_0 \star j_1 : Q_R A \to A \oplus A.$$

We will show that the continuous algebra homomorphism

$$j : A \oplus A \longrightarrow M_2(Q_R A), \quad (a, b) \longmapsto \begin{pmatrix} \iota_0(a) & 0 \\ 0 & \iota_1(b) \end{pmatrix}$$

defines an inverse of i modulo smooth-homotopy, after stabilization by 2×2-matrices. We compute for $(a, b) \in A \oplus A$

$$(M_2(i) \circ j)(a, b) = \begin{pmatrix} i(\iota_0(a)) & 0 \\ 0 & i(\iota_1(b)) \end{pmatrix} = \begin{pmatrix} (a, 0) & 0 \\ 0 & (0, b) \end{pmatrix}.$$

Since the following assignment defines a smooth function

$$t \in [0, 1] \longmapsto u_t := \begin{pmatrix} \cos(\frac{\pi}{2}t) & \sin(\frac{\pi}{2}t) \\ -\sin(\frac{\pi}{2}t) & \cos(\frac{\pi}{2}t) \end{pmatrix} \in M_2(A \oplus A),$$

the family of homomorphisms $\{h_t\}_t$ given by

$$h_t : A \oplus A \longrightarrow M_2(A \oplus A), \quad (a, b) \longmapsto \begin{pmatrix} (a, 0) & 0 \\ 0 & 0 \end{pmatrix} + u_t \begin{pmatrix} 0 & 0 \\ 0 & (0, b) \end{pmatrix} u_t^{-1}$$

realizes a smooth homotopy between $h_0 = M_2(i) \circ j$ and the top-left corner inclusion $h_1 : A \oplus A \hookrightarrow M_2(A \oplus A)$. Now the other side composition gives for $a \in A$

$$(j \circ i)(\iota_0(a)) = j(a, 0) = \begin{pmatrix} \iota_0(a) & 0 \\ 0 & 0 \end{pmatrix} \text{ and } (j \circ i)(\iota_1(a)) = j(a, 0) = \begin{pmatrix} 0 & 0 \\ 0 & \iota_1(a) \end{pmatrix}.$$

For $t \in [0,1]$, write $P_0^t := E_{11}$ and $P_1^t := u_t E_{22} u_t^{-1} \in M_2(\mathbb{C})$ for elementary matrices conjugated by the rotation u_t. It is clear that the assignment $t \mapsto P_0^t, P_1^t$ define smooth function on the interval. Consider the algebra homomorphisms

$$\gamma_0^t : A \longrightarrow M_2(Q_R A), \quad a \longmapsto P_0^t \iota_0(a),$$

$$\gamma_1^t : A \longrightarrow M_2(Q_R A), \quad a \longmapsto P_1^t \iota_1(a).$$

Their free product $h^t := \gamma_0^t \star \gamma_1^t : Q_R A \to M_2(Q_R A)$ is continuous for the respective Fréchet topology. Indeed, if $x \in A \star A$ is a word of $n+1$-length then

$$\begin{aligned} \|(\gamma_0^t \star \gamma_1^t)(x)\|_{(R,N)} &= \|(P_0^t \cdots P_{n \bmod 2}^t)(x)\|_{(R,N)} \\ &\leq \|P_0^t \cdots P_{n \bmod 2}^t\|_{(R,N)} \|x\|_{(R,N)} \\ &\leq 4\|x\|_{(R,N)} \end{aligned}$$

because the product $P_0^t \cdots P_{n \bmod 2}^t \in M_2(\mathbb{C})$ is a scalar matrix where each coefficient is at most 1. It is clear that the family $\{h^t\}_t$ depends smoothly on the parameter $t \in [0,1]$ and that it realizes an homotopy between $h^0 = j \circ i$ and the top-left corner inclusion $h^1 : Q_R A \hookrightarrow M_2(Q_R A)$. We showed that the algebra homomorphism $i : Q_R A \to A \oplus A$ is invertible modulo smooth-homotopy after stabilization by 2×2-matrices, of inverse given by $j : A \oplus A \to M_2(Q_R A)$.
Finally, suppose a real Lie group G acts smoothly on the Fréchet algebra A in a way such that the induced action is also smooth on $Q_R A$. By definition, the action commutes with the canonical inclusions $\iota_{0,1}$ and the rotation matrices u_t, thus commutes with the smooth homotopies h_t and h^t. It ends up the proof. □

The rescaled Cuntz' algebra

Definition 4.3 For all $R \geq 1$, the **rescaled Cuntz's algebra** $q_R A$ associated to a Banach algebra A is the kernel of the continuous fold map $id \star id : Q_R A \to A$ arising from the pair of identity morphisms

$$q_R A := \ker(id \star id : Q_R A \to A).$$

It fits into the following splited extension of algebras endowed with two bounded linear sections

$$0 \longrightarrow q_R A \hookrightarrow Q_R A \underset{\iota_0,\, \iota_1}{\overset{id \star id}{\rightleftarrows}} A \longrightarrow 0 \ . \tag{19}$$

The next lemma shows that the rescaled Cuntz' algebra $q_R A$ is the *completion of the two-sided ideal of differences* in $Q_R A$. Indeed, it shows that the reduced words of $q_R A$ belong in the two-sided ideal of differences in $Q_R A$ while any difference $\iota_0(a) - \iota_1(a)$ with $a \in A$ belongs to $q_R A$ by the universal property of the free product.

Lemma 4.4. *Any reduced word of the form $x = \iota_0(a_0)\iota_1(a_1) \cdots \iota_{n \bmod 2}(a_n) \in q_R A$ with $a_i \in A$ can be written as:*

$$x = \sum_{k=0}^{\lfloor n/2 \rfloor} \iota_1(a_0 \cdots a_{2k-1}) \Big(\iota_0(a_{2k}) - \iota_1(a_{2k}) \Big) \iota_1(a_{2k+1}) \iota_0(a_{2k+1}) \cdots \iota_{n \bmod 2}(a_n).$$

Proof. A straightforward computation shows that telescopic sum above is equal to $x - \iota_1(a_0 \cdots a_n) \in Q_R A$. But because x is in the kernel of the fold map,

$$0 = (id \star id)(x) = (id \star id)(\iota_0(a_0)\iota_1(a_1) \cdots \iota_{n \bmod 2}(a_n)) = a_0 \cdots a_n \in A.$$

Then the sum above is exactly equal to x and we showed the lemma. □

The following theorem is central in the study. It relies on the fact that the homotopy equivalence between the free product $Q_R A$ and $A \oplus A$ described in 4.2 is stable enough to be preserved under smooth actions of group and tensorization. It is a version of the constructions of [Cun87] cited in theorem 3.12 for Fréchet algebras.

Theorem 4.5. *Let A be a Fréchet algebra and $\mathcal{A}$ a locally convex algebra. Suppose they are both endowed with a smooth action of a real Lie group G and that the G-action on A extends as a smooth G-action on $Q_R A$. For any $R \geq 0$, the algebra homomorphism $id \star 0 : q_R A \longrightarrow A$ induces a continuous chain-homotopy equivalence*

$$\widehat{CC}(((id \star 0) \otimes_\pi id_{\mathcal{A}}) \rtimes G) : \widehat{CC}((q_R A \otimes_\pi \mathcal{A}) \rtimes G) \xrightarrow{\sim} \widehat{CC}((A \otimes_\pi \mathcal{A}) \rtimes G).$$

Proof. Consider the extension $0 \longrightarrow q_R A \longrightarrow Q_R A \longrightarrow A \longrightarrow 0$ as in (19) with the two bounded linear sections ι_0 and ι_1. Since it is split, it remains exact and split once we take the projective tensor product with the locally convex algebra $\mathcal{A}$ and the crossed product by the real Lie group G. Namely, the extension

$$0 \longrightarrow (q_R A \otimes_\pi \mathcal{A}) \rtimes G \longrightarrow (Q_R A \otimes_\pi \mathcal{A}) \rtimes G \xrightarrow{id \star id} (A \otimes_\pi \mathcal{A}) \rtimes G \longrightarrow 0$$

is exact and endowed with bounded linear sections. Excision in periodic cyclic homology 1.3 shows that there exists a continuous chain-homotopy equivalence

$$\widehat{CC}((q_R A \otimes_\pi \mathcal{A}) \rtimes G) \xrightarrow{\sim} \mathrm{Cone}\Big(\widehat{CC}((Q_R A \otimes_\pi \mathcal{A}) \rtimes G) \xrightarrow{id \star id} \widehat{CC}((A \otimes_\pi \mathcal{A}) \rtimes G)\Big).$$

Moreover, the theorem 4.2 shows that the algebra homomorphism

$$(i \otimes id_{\mathcal{A}}) \rtimes G : (Q_R A \otimes_\pi \mathcal{A}) \rtimes G \longrightarrow [(A \otimes_\pi \mathcal{A}) \rtimes G] \oplus [(A \otimes_\pi \mathcal{A}) \rtimes G]$$

is invertible modulo stabilization by 2×2-matrices, namely induces a continuous chain-homotopy equivalence

$$\widehat{CC}((A \otimes_\pi \mathcal{A}) \rtimes G) \xrightarrow{\sim} \mathrm{Cone}\Big(\widehat{CC}((Q_R A \otimes_\pi \mathcal{A}) \rtimes G) \longrightarrow \widehat{CC}((A \otimes_\pi \mathcal{A}) \rtimes G)\Big).$$

The result follows from the identification of the left-hand terms by the continuous algebra homomorphism $id \star 0 : q_R A \longrightarrow A$. □

Corollary 4.6. *Let A and A' two Fréchet algebras and $\mathcal{A}$ a locally convex algebra, all endowed with a smooth action of a real Lie group G which extends smoothly on their free algebras. For any $R > 0$, the morphisms:*

$$\Delta_R = \Big(\Big(\iota_0^A \otimes id\Big) \star \Big(\iota_1^A \otimes id\Big)\Big)_{|q_R} : q_R(A \otimes_\pi A') \longrightarrow q_R A \otimes_\pi A',$$

$$\nabla_R = \Big(\Big(id \otimes \iota_0^{A'}\Big) \star \Big(id \otimes \iota_1^{A'}\Big)\Big)_{|q_R} : q_R(A \otimes_\pi A') \longrightarrow A \otimes_\pi q_R A'$$

induce chain-homotopy equivalences trough the functor $\widehat{CC}((- \otimes_\pi \mathcal{A}) \rtimes G)$.

Proof. It is a consequence of the identities

$$((id_A \star 0) \otimes id_{A'}) \circ \Delta_R = id_{A\otimes_\pi A'} \star 0 \quad \text{and} \quad (id_A \otimes (id_{A'} \star 0)) \circ \nabla_R = id_{A\otimes_\pi A'} \star 0$$

and the fact that $id \star 0$ is a chain-homotopy equivalences through the functor $\widehat{CC}((- \otimes_\pi \mathcal{A}) \rtimes G)$ due to the previous theorem. □

Recall that the Riemannian connection ∇ acts as a derivation on the smooth Clifford module $\mathcal{C}^\infty_\tau(G/K)$. We endowed this algebra with Fréchet topology (11) coming from the family of $\mathcal{C}^k$-sup norms $\|f\|_k := \|\nabla^k(f)\|$ with $k \in \mathbb{N}$. Here, the smooth action of G by left-translation induces, for any $R \geq 0$, a smooth action on the free algebra $Q_R(\mathcal{C}^\infty_\tau(G/K))$ because $\nabla^k(g \cdot f) = g \cdot \nabla^k(f)$ and hence preserves the norms. The following corollary is obtained a direct consequence of the theorems 3.4 and 4.5.

Corollary 4.7. *Let $\mathcal{A}$ be a locally convex unital algebra endowed with a smooth action of a real Lie group G. For any $R \geq 0$, there exists a isomorphism in periodic cyclic homology*

$$HP_\bullet(\mathcal{C}^\infty_c(G/K, \mathcal{A}) \rtimes G) \xrightarrow{\sim} HP_\bullet((q_R(\mathcal{C}^\infty_\tau(G/K)) \otimes_\pi \mathcal{A}) \rtimes G).$$

4.2 Fréchet version of Dirac and dual Dirac elements

We recall that the smooth Clifford algebra $\mathcal{C}^\infty_\tau(G/K)$ is Fréchet for the topology described in (11). Thus one can consider, for any $R \geq 1$, its rescaled Cuntz algebra $q_R(\mathcal{C}^\infty_\tau(G/K))$. In this subsection, we verify that there exists a parameter $R \geq 1$ great enough so that the Dirac and dual Dirac elements descend to respectively G-equivariant and K-equivariant morphisms of Fréchet algebras

$$\alpha^\sharp : q_R(\mathcal{C}^\infty_\tau(G/K)) \longrightarrow \ell^{p,G}(\mathcal{H}) \quad \text{and} \quad \beta^\sharp : q_1\mathbb{C} \longrightarrow \mathcal{C}^\infty_\tau(G/K) \otimes_\pi M_2(\mathbb{C})$$

where $\ell^{p,G}(\mathcal{H})$ is the G-smooth p-th Schatten ideal (defined in (16)) of the infinite dimensional Hilbert space $\mathcal{H} = L^2(\Omega^\star(G/K))$. We will also compute their composition $\alpha^\sharp \circ q\beta^\sharp$ which turns out to be K-equivariantly homotopic to the morphism $\mathbb{1}_K$ representing the trivial class $1 \in KK_K(\mathbb{C}, \mathbb{C})$ (see 3.15).

Recall that the Dirac element $\alpha = [(\rho, F, U)]$ defines a class in G-equivariant bivariant K-theory 3.7 with $\mathcal{H} = L^2(\Omega^\star(G/K))$ as underlying Hilbert space. Since G/K is supposed to be even-dimensional, we decompose $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$ into even and odd part and denote P_+ for the projection onto the even part. Consider the continuous $\star$-morphisms

$$\rho_0 := P_+\rho P_+ : \mathcal{C}^\infty_\tau(G/K) \longrightarrow \mathcal{L}(\mathcal{H}) \quad \text{and} \quad \rho_1 := P_+F\rho F P_+ : \mathcal{C}^\infty_\tau(G/K) \longrightarrow \mathcal{L}(\mathcal{H}).$$

Proposition 4.8. *The Dirac element $\alpha \in KK_G(\mathcal{C}_\tau(G/K), \mathbb{C})$ can be represented by a G-equivariant Kasparov bimodule (ρ, F, U) such that for $p > \dim(G/K)$ there exists a constant $R > 0$ verifying for all $f \in \mathcal{C}^\infty_\tau(G/K)$:*

$$F^2 = 1, \quad \|\rho_i(f)\| \leq R\|f\|_0 \quad \textit{and} \quad \|\rho_0(f) - \rho_1(f)\|_{\ell^p(\mathcal{H})} \leq R\|f\|_1.$$

where $\|f\|_0$ and $\|f\|_1$ denote the sup-norms of the function f and its derivative, defined in (11).

Proof. According to Connes [Con85, pages 89-91] one may obtain, after adding a degenerate Kasparov bimodule to α, a G-invariant Fredholm module (ρ, F, U) which represents the same class as α in the group $KK_G(C_\tau(G/K), \mathbb{C})$, is p-summable over $\mathcal{C}_c^\infty(G/K)$ for a certain $p > \dim(G/K)$ and satisfies $F^2 = Id$.
It is clear that the morphisms ρ_0 and ρ_1 are bounded because P_+, ρ and F are also by definition of a Fredholm module. We have to show that their difference is bounded for the ℓ^p-norm. An easy computation gives

$$\rho_0(f) - \rho_1(f) = P_+ F [F, \rho(f)] P_+.$$

The operator P_+ and F being ℓ^p-bounded using the refinement of Connes, it remains to show that the commutator $[F, \rho(f)]$ is also. Let $x \in G/K$ and $B_1 \subset B_2 \subset B_3 \subset G/K$ be balls of center x and strictly increasing radii. We will show that $[F, \rho(f)]$ is ℓ^p-bounded for any $f \in \mathcal{C}_c^\infty(B_1)$. Fix a real cutoff function $\chi \in \mathcal{C}_c^\infty(B_2)$ which is constant equal to 1 on B_1 and $\mu \in \mathcal{C}_c^\infty(B_3)$ which is constant equal to 1 on B_2. The commutator becomes:

$$[F, \rho(f)] = [F, \rho(f)]\chi + \rho(f)[F, \chi].$$

The image of the operator $\rho(f)[F, \chi]$ consists of functions supported in B_1. Viewing them as functions on $S^n \approx G/K \cup \{\infty\}$ vanishing outside B_1 we may interprete $\rho(f)[F, \chi]$ as pseudodifferential operator from G/K to S^n. Its order is -1 so that it extends to a bounded operator $\rho(f)[F, \chi] : \mathcal{L}^2(G/K) \longrightarrow \mathcal{H}^{(1)}(S^n)$ whose norm is majorized in terms of f, χ and their first derivatives, i.e. in terms of $\|f\|_1$. Now we may factorize

$$\rho(f)[F, \chi] : \mathcal{L}^2(G/K) \longrightarrow \mathcal{H}^{(1)}(S^n) \longrightarrow \mathcal{L}^2(S^n) \xrightarrow{\cdot\mu} \mathcal{L}^2(G/K)$$

where the last map is given by multiplication with μ, followed by extension by 0 outside B_3. The first and last maps are bounded operators whereas the middle one lies in the p-th Schatten ideal for p great enough. The assertion for $\rho(f)[F, \chi]$ follows because ℓ^p is a two-sided ideal in $\mathcal{L}(H)$. Consider now the bounded operator $[F, \rho(f)]\chi$ on $\mathcal{L}^2(G/K)$. The operator F is bounded and self-adjoint on $\mathcal{L}^2(G/K)$. Now a bounded operator on Hilbert space belongs to the Schatten ideal ℓ^p if and only if its adjoint does and the Schatten norms of the two operators coincide. We find

$$([F, \rho(f)]\chi)^\star = -\chi[F, \rho(f)]$$

and our previous arguments apply to the latter operator. This proves that $[F, \rho(f)]$ is bounded for the ℓ^p-norm. Choose R great enough with respect to the norm of the ρ_i and of the commutator $[F, \rho(f)]$. □

Theorem 4.9. *The Dirac element α induces, for $p > \dim(G/K)$ and $R > 0$ great enough, a continuous G-equivariant morphism of Fréchet algebras*

$$\alpha^\sharp = (\rho_0 \star \rho_1)_{|q_R} : q_R(\mathcal{C}_\tau^\infty(G/K)) \longrightarrow \ell^{p,G}(\mathcal{H}).$$

where we recall that $\ell^{p,G}(\mathcal{H})$ is the G-smooth p-th Schatten ideal (definition 16).

Proof. Take R as in the lemma above. Due to the Fréchet topology on $q_R(\mathcal{C}_\tau^\infty(G/K))$ (see (11) and (18)) coming from the $\mathcal{C}^k$-sup norms on $\mathcal{C}_\tau^\infty(G/K)$, it suffices to show

that for any reduced word $x = \iota_0(f_0)\iota_1(f_1)\cdots\iota_{n \bmod 2}(f_n) \in q_R(\mathcal{C}^\infty_\tau(G/K))$ with $f_i \in \mathcal{C}^\infty_\tau(G/K)$, it exists a constant $C > 0$ and an integer $m \in \mathbb{N}$ such that

$$\|(\rho_0 \star \rho_1)(x)\|_{\ell^p(H)} \leq C\|x\|_{(R,m)}.$$

Due to the lemma 4.4, we can write

$$(\rho_0 \star \rho_1)(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \rho_1(f_0 \cdots f_{2k-1})\Big(\rho_0(f_{2k}) - \rho_1(f_{2k})\Big)\rho_1(f_{2k+1})\cdots\rho_{n \bmod 2}(f_n).$$

The free product $\rho_0 \star \rho_1$ applied to $x \in q_R A$ is of norm:

$$\begin{aligned}
\|(\rho_0 \star \rho_1)(x)\|_{\ell^p(H)} &\leq \sum_{k=0}^{\lfloor n/2 \rfloor} \|\rho_1(f_0 \cdots f_{2k-1})\Big(\rho_0(f_{2k}) - \rho_1(f_{2k})\Big)\rho_1(f_{2k+1})\cdots\rho_{n \bmod 2}(f_n)\|_{\ell^p(H)} \\
&\leq \sum_{k=0}^{\lfloor n/2 \rfloor} \|\rho_1(f_0 \cdots f_{2k-1})\|_{L(H)}\|\rho_0(f_{2k}) - \rho_1(f_{2k})\|_{\ell^p(H)}\|\rho_1(f_{2k+1})\cdots\rho_{n \bmod 2}(f_n)\|_{L(H)} \\
&\leq \sum_{k=0}^{\lfloor n/2 \rfloor} R^n\|f_0\|_0 \cdots \|f_{2k-1}\|_0\|\rho_0(f_{2k}) - \rho_1(f_{2k})\|_{\ell^p(H)}\|f_{2k+1}\|_0 \cdots \|f_n\|_0 \\
&\qquad\qquad \text{(by 4.8)} \\
&\leq \sum_{k=0}^{\lfloor n/2 \rfloor} R^{n+1}\|f_0\|_0 \cdots \|f_{2k-1}\|_0\|f_{2k}\|_1\|f_{2k+1}\|_0 \cdots \|f_n\|_0 \qquad \text{(by 4.8)} \\
&\leq R\|x\|_{(R,1)},
\end{aligned}$$

which ends up the proof of boundness. Finally, the image of $\rho_0 \star \rho_1$ is contained in the space of smooth vectors of the G-module $\ell^p(\mathcal{H})$ because F is G-equivariant and acts smoothly on $\mathcal{C}^\infty_\tau(G/K)$. □

Proposition 4.10. *The dual Dirac element $\beta = [e_0] - [e_1] \in KK_K(\mathbb{C}, \mathcal{C}_\tau(G/K))$ induces a K-equivariant continuous morphism of Fréchet algebras*

$$\beta^\sharp = (e_0 \star e_1)_{|q_1\mathbb{C}} : q_1\mathbb{C} \longrightarrow M_2(\mathcal{C}^\infty_\tau(G/K)).$$

Proof. It is a consequence of the fact that the idempotent 2×2-matrices e_0 and e_1 are K-equivariant and of norm 1. Their difference is compactly-supported as shown in 3.9. □

Proposition 4.11. *The following continuous algebra homomorphisms are K-equivariantly smoothly homotopic*

$$\alpha^\sharp \circ q_R\beta^\sharp : q_R(q_1\mathbb{C}) \longrightarrow M_2(\ell^{p,G}(\mathcal{H})) \quad \textit{and} \quad \mathbb{1}_K : q_R(q_1\mathbb{C}) \longrightarrow M_2(\ell^{p,G}(\mathcal{H})).$$

Proof. According to the lemma (17), it suffices to show that the following compositions $\alpha^\sharp \circ q_R\beta^\sharp$ and $\alpha^{\mathrm{Euc}} \circ q_R\beta^{\mathrm{Euc}}$ are smoothly homotopy equivalent. Consider a smooth partition of unity $\{\chi_i\}_{i\in I}$ of G/K which is locally finite, verifies $\sum_i \chi_i^2 = 1$, and whose diameters of supports $\mathrm{diam}(\mathrm{supp}(\chi_i)) \leq r$ is bounded by a fixed real number $r > 0$. Then, the operator

$$Q := \sum_{i\in I} \chi_i(1 + D^2)^{-1/2}\chi_i$$

is of finite-propagation at most r and defines a parametrix of $D := d + d^\star$. By convexity of the set of parametrix, there exists an affine (smooth) homotopy between

$F = D(1+D^2)^{-1/2}$ and $F' = DQ$. Also, under the identification $\mathbb{R}^q \sim G/K$, the affine homotopy of Riemann metrics $t \mapsto (1-t)\boldsymbol{g_0} + t\boldsymbol{g_1}$, where $\boldsymbol{g_0}$ is a Riemann metrix on $\mathbb{R}^q$ and $\boldsymbol{g_1}$ on G/K, provides a smooth homotopy between F' and the bounded Dirac operator on $\mathbb{R}^q$. Namely, the composition of the affine homotopy of parametrices and the affine homotopy of the Riemann metrix together provides a smooth homotopy between $\alpha^\sharp$ and α^{Euc}. Since we chose a parametrix with finite propagation, the support of $e_0 - e_1$ remains compact under this composition of homotopies, and then provides a smooth homotopy between $\beta^\sharp$ on G/K and its Euclidian version β^{Euc}, which ends up the proof. □

Remark We expected these versions of the Dirac and dual Dirac elements in periodic cyclic homology to be K-equivariantly inverse to each others, as for the Dirac and dual Dirac elements in bivariant K-theory. Unfortunately, the last proposition states that one side of their composition is not the identity morphism but a representative of the trivial class in $KK_K(\mathbb{C},\mathbb{C})$. Indeed, we don't know if the morphism $\mathbb{1}_K$ realizes the identity morphism in periodic cyclic homology because we are unable to compute the periodic cyclic homology of its codomain $\ell^{p,G}(\mathcal{H})$. We are able to prove that the p-th Schatten ideal $\ell^p(\mathcal{H})$ is quasi-Morita equivalent to $\ell^1(\mathcal{H})$ for any $p > 1$ and that $\ell^1(\mathcal{H})$ possesses the same periodic cyclic homology than the complex field $\mathbb{C}$, but we still don't know anything about its subalgebra of G-smooth vectors.

In the next section we will define a stabilization algebra $\mathcal{D}$ so that when we take the tensor product by $\mathcal{D}$, the algebra morphism $\mathbb{1}_K$ becomes the identity. This algebra contains the degeneracy of the Dirac and dual Dirac in periodic cyclic homology to be inverse to each others. This is the reason why our main result is established *after stabilization.*

4.3 The stabilization algebra

Consider the infinite dimensional Hilbert space $\mathcal{H} = L^2(\Omega^\star(G/K)) \oplus \mathbb{C}$, the L^2-completion of the compactly supported differential forms $\Omega^\star_c(G/K)$, with an extra copy of $\mathbb{C}$. It is endowed with the G-action by translations on G/K and the trivial action on $\mathbb{C}$. The rank-one projection $P : \mathcal{H}\oplus\mathbb{C} \longrightarrow\!\!\!\!\!\rightarrow \mathbb{C}$ is isometric and G-smooth, namely $P \in \ell^{p,G}(\mathcal{H})$ for any integer $p \geq 1$. Since its image is made of K-fixed vectors the lemma 3.15 states that the following morphism represents the class of $1 \in KK_K(\mathbb{C},\mathbb{C})$:

$$\begin{array}{rccc} \mathbb{1}_K : q(q\mathbb{C}) & \xrightarrow{(id\star 0)\circ(id\star 0)} & \mathbb{C} & \xrightarrow{\quad\iota\quad} & M_2(\ell^{p,G}(\mathcal{H})) \\ & & 1 & \longmapsto & \begin{pmatrix} P & 0 \\ 0 & 0 \end{pmatrix} \end{array} . \tag{20}$$

The projection P being isometric, the system of continuous inclusions of algebras

$$\begin{array}{ccc} \ell^{p,G}(\mathcal{H})^{\otimes_\pi n} & \hookrightarrow & \ell^{p,G}(\mathcal{H})^{\otimes_\pi n+1} \\ x & \longmapsto & P\otimes x \end{array} \tag{21}$$

is compatible with the successive norms.

Definition 4.12 The **stabilization algebra** $\mathcal{D}$ is defined to be the completion of the inductive limit of the system above, for the inductive norm

$$\mathcal{D} := \overline{\lim_{\to n} \ell^{p,G}(\mathcal{H})^{\otimes_\pi n}}.$$

Given any locally convex algebra A endowed with a smooth action the almost-connect real Lie group G, we will write $\mathcal{A} := A \otimes_\pi \mathcal{D}$ for its **stabilized algebra**. Since $\mathcal{D}$ is locally convex and endowed with a smooth action of G, the stabilization $\mathcal{A}$ is also locally convex and endowed with the diagonal smooth action of G.

Lemma 4.13. *For every locally convex algebra A endowed with a smooth action of G, with stabilization $\mathcal{A} = A \otimes_\pi \mathcal{D}$, the inclusion $(1 \mapsto P) : \mathbb{C} \hookrightarrow \ell^{p,G}(\mathcal{H})$ induces an isomorphism in periodic cyclic homology*

$$((x \mapsto P \otimes x) \otimes id_{\mathcal{A}}) \rtimes G : HP_\bullet(\mathcal{A} \rtimes G) \xrightarrow{\sim} HP_\bullet\Big((\ell^{p,G}(\mathcal{H}) \otimes_\pi \mathcal{A}) \rtimes G\Big).$$

Proof. The stabilization algebra $\mathcal{D}$ is naturally isomorphic to $\ell^{p,G}(\mathcal{H}) \otimes_\pi \mathcal{D}$ by definition of the inductive limit and under this identification, the shift morphism (21) from $\mathcal{D}$ to $\ell^{p,G}(\mathcal{H}) \otimes_\pi \mathcal{D} \simeq \mathcal{D}$ is equal to the identity of $\mathcal{D}$. In particular, after tensoring with A, we also get the identification $\ell^{p,G}(\mathcal{H}) \otimes_\pi \mathcal{A} \simeq \mathcal{A}$ and that $(x \mapsto P \otimes x) \otimes id_{\mathcal{A}}$ is equal to $id_{\mathcal{A}}$. Everything is compatible with the action of G as P is G-smooth. □

Corollary 4.14. *For every locally convex algebra A endowed with a smooth action of G, with stabilization $\mathcal{A} = A \otimes_\pi \mathcal{D}$, the morphism $\mathbb{1}_K$ (see (20)) representing the trivial class in $KK_K(\mathbb{C}, \mathbb{C})$ induces an isomorphism in periodic cyclic homology*

$$(\mathbb{1}_K \otimes id_{\mathcal{A}}) \rtimes G : HP_\bullet((q_R(q_1\mathbb{C}) \otimes_\pi \mathcal{A}) \rtimes G) \xrightarrow{\sim} HP_\bullet\Big((M_2(\ell^{p,G}(\mathcal{H})) \otimes_\pi \mathcal{A}) \rtimes G\Big).$$

Proof. By the previous lemma, the morphism $\iota \otimes id_{\mathcal{A}} : \mathcal{A} \longrightarrow M_2(\ell^{p,G}(\mathcal{H})) \otimes_\pi \mathcal{A} \simeq M_2(\mathcal{A})$ is nothing other than the top-left inclusion of $\mathcal{A}$ in its 2×2-matrices algebra. We use theorem 4.5 and the Morita invariant of periodic cyclic homology 1.3 to conclude. Again, everything is compatible with the action of G as P is G-smooth. □

The next corollary shows that the stabilization algebra plays the expected role. Namely, after tensorization by this algebra, the Dirac element $\alpha^\sharp$ realizes the linear map (9) and the dual Dirac element $\beta^\sharp$ provides a K-equivariant right-inverse of it.

Corollary 4.15. *For any locally convex unital algebra A endowed with a smooth action G and of stabilization algebra $\mathcal{A} = A \otimes_\pi \mathcal{D}$,*

1. *the G-equivariant morphism of Fréchet algebras $\alpha^\sharp$ induces a linear map*

$$HP((\alpha^\sharp \otimes id_{\mathcal{A}}) \rtimes G) : HP_\bullet(\mathcal{C}_c^\infty(G/K, \mathcal{A}) \rtimes G) \longrightarrow HP_\bullet(\mathcal{A} \rtimes G);$$

2. *the K-equivariant morphism of Fréchet algebras $\beta^\sharp$ is such that $HP((\beta^\sharp \otimes id_{\mathcal{A}}) \rtimes K)$ realizes a right-inverse of $HP((\alpha^\sharp \otimes id_{\mathcal{A}}) \rtimes K)$.*

Proof. 1. Recall that the Dirac element α induces a G-equivariant morphism of Fréchet algebras $\alpha^\sharp : q_R(\mathcal{C}^\infty_\tau(G/K)) \longrightarrow \ell^{p,G}(\mathcal{H})$ by theorem 4.9. Using the isomorphisms of corollaries 4.7 and 4.13 we obtain the following composition

$$\begin{array}{ccc} HP_\bullet\Big(\big(q_R(\mathcal{C}^\infty_\tau(G/K))\otimes_\pi \mathcal{A}\big)\rtimes G\Big) & \xrightarrow{(\alpha^\sharp\otimes_\pi id_{\mathcal{A}})\rtimes G} & HP_\bullet\Big(\big(\ell^{p,G}(\mathcal{H})\otimes_\pi \mathcal{A}\big)\rtimes G\Big) \\ \sim\uparrow & & \uparrow\sim \\ HP_\bullet(\mathcal{C}^\infty_c(G/K,\mathcal{A})\rtimes G) & & HP_\bullet(\mathcal{A}\rtimes G) \end{array}$$

2. The proposition 4.11 states that there exists a K-equivariant smooth homotopy equivalence between the morphisms of algebras

$$\alpha^\sharp\circ q_R\beta^\sharp : q_R(q_1\mathbb{C}) \longrightarrow M_2(\ell^{p,G}(\mathcal{H})) \text{ and } \mathbb{1}_K = \iota\circ(id\star 0)^2 : q_R(q_1\mathbb{C}) \longrightarrow M_2(\ell^{p,G}(\mathcal{H})).$$

The second homomorphism induces an isomorphism in periodic cyclic homology through the functor $HP_\bullet((-\otimes_\pi \mathcal{A})\rtimes G)$ by the corollary above, and so is the first. The linear map $HP((q_R\beta^\sharp\otimes id_{\mathcal{A}})\rtimes K)$ then realizes a right-inverse of $HP((\alpha^\sharp\otimes id_{\mathcal{A}})\rtimes K)$. Finally, $HP((q_R\beta^\sharp\otimes id_{\mathcal{A}})\rtimes K) = HP((\beta^\sharp\otimes id_{\mathcal{A}})\rtimes K)$ by theorem 4.5. □

4.4 Rotation trick

In the previous section we established that the dual Dirac element provides a K-equivariant rigt-inverse of the Dirac element, so we are now interested in the left-invertibility of the Dirac element. To do so, we will use an analogue of Atiyah's rotation trick to compute a side of their composition using their other-side composition. At the bivariant K-theory level, this has to be thought as the computation of the Kasparov product $\alpha\times\beta$ using the other-side product $\beta\times\alpha$. We will realize this rotation trick using the flip of coordinates in the Euclidian manifold $G/K\times G/K$. **For convenience, we will write $\mathcal{M}$ for the algebra $M_2(\mathcal{C}^\infty_\tau(G/K))$.**

Lemma 4.16 (Rotation trick)**.** *Recall the definition of the dual Dirac element β using the idempotent matrices $e_0, e_1 \in \mathcal{M}$ (3.9). The rotation that flips the coordinates in $G/K\times G/K$ realizes a K-equivariant smooth homotopy between the following morphisms, for $i = 0, 1$:*

$$e_i\otimes id : \mathcal{M}\longrightarrow \mathcal{M}\otimes_\pi\mathcal{M} \quad \textit{and} \quad id\otimes e_i : \mathcal{M}\longrightarrow\mathcal{M}\otimes_\pi\mathcal{M}.$$

Proof. Since by assumption G/K is even dimensional, the smooth Clifford module splits into even and odd parts $\mathcal{C}^\infty_\tau(G/K) = \mathcal{C}^\infty_\tau(G/K)_+\oplus\mathcal{C}^\infty_\tau(G/K)_-$ as in the equation (10). We write ε for the grading automorphism which is the identity on its even part and minus the identity on its odd part. A direct computation shows that for any $x\in\mathbb{R}^q$ and $i = 0, 1$ we have $\varepsilon(e_i(-x)) = e_i(x)$. Namely, it suffices to show that the following algebra homomorphism is smoothly homotopic to the identity:

$$\begin{array}{rccc} \Theta: & \mathcal{C}^\infty_\tau(G/K)\otimes_\pi\mathcal{C}^\infty_\tau(G/K) & \longrightarrow & \mathcal{C}^\infty_\tau(G/K)\otimes_\pi\mathcal{C}^\infty_\tau(G/K) \\ & a\otimes b & \longmapsto & (x,y)\mapsto b(x)\otimes\varepsilon(a(-y)) \end{array}.$$

Recall the identification

$$\mathcal{C}^\infty_\tau(G/K)\otimes_\pi\mathcal{C}^\infty_\tau(G/K)\simeq\mathcal{C}^\infty_c(G/K\times G/K,\mathrm{Cl}_{\mathbb{C}}(\mathbb{R}^q\oplus\mathbb{R}^q)).$$

The smooth family of rotations for $0 \leq t \leq \pi/2$

$$h_t : G/K \times G/K \longrightarrow G/K \times G/K,\ (v, v') \mapsto (\cos(t)v + \sin(t)v',\ -\sin(t)v + \cos(t)v')$$

joins the map $(x, y) \mapsto (-y, x)$ to the identity and the smooth family of linear isometries for $0 \leq t \leq \pi/2$

$$h'_t : \mathbb{R}^q \oplus \mathbb{R}^q \longrightarrow \mathbb{R}^q \oplus \mathbb{R}^q,\ (v, v') \mapsto (\cos(t)v + \sin(t)v',\ -\sin(t)v + \cos(t)v')$$

induce a smooth homotopy of algebra automorphisms $\varphi_t : \mathrm{Cl}_{\mathbb{C}}(\mathbb{R}^q \oplus \mathbb{R}^q) \longrightarrow \mathrm{Cl}_{\mathbb{C}}(\mathbb{R}^q \oplus \mathbb{R}^q)$ between $\varphi_0 = id$ and $\varphi_{\pi/2}(a \otimes b) = b \otimes \varepsilon(a)$. The composition $\varphi_t \circ \Theta \circ h_t$, $0 \leq t \leq \frac{\pi}{2}$ provides the expected smooth homotopy. It is K-equivariant because the homeomorphism $G/K \sim \mathbb{R}^q$ transports the K-action to the action by rotations on $\mathbb{R}^q$. □

Proposition 4.17. *The following algebra morphisms are chain-homotopy equivalent through the functor $\widehat{CC}((- \otimes_\pi \mathcal{A}) \rtimes K)$ for any locally convex algebra $\mathcal{A}$ endowed with a smooth action of K:*

$$\beta^\sharp \otimes id_{\mathcal{M}} : q_1\mathbb{C} \otimes_\pi \mathcal{M} \longrightarrow \mathcal{M} \otimes_\pi \mathcal{M} \quad \textit{and} \quad id_{\mathcal{M}} \otimes \beta^\sharp : \mathcal{M} \otimes_\pi q_1\mathbb{C} \longrightarrow \mathcal{M} \otimes_\pi \mathcal{M}$$

Proof. By definition of Δ and ∇ in 4.6, it is clear that the following diagrams commute

$$\begin{array}{ccc} q_1\mathcal{M} & \xrightarrow{(e_0 \otimes id) \star (e_1 \otimes id)} & \mathcal{M} \otimes_\pi \mathcal{M} \\ {\scriptstyle \Delta_1}\downarrow & & \| \\ q_1\mathbb{C} \otimes_\pi \mathcal{M} & \xrightarrow{\beta^\sharp \otimes id_{\mathcal{M}}} & \mathcal{M} \otimes_\pi \mathcal{M} \end{array} \qquad \begin{array}{ccc} q_1\mathcal{M} & \xrightarrow{(id \otimes e_0) \star (id \otimes e_1)} & \mathcal{M} \otimes_\pi \mathcal{M} \\ {\scriptstyle \nabla_1}\downarrow & & \| \\ \mathcal{M} \otimes_\pi q_1\mathbb{C} & \xrightarrow{id_{\mathcal{M}} \otimes \beta^\sharp} & \mathcal{M} \otimes_\pi \mathcal{M} \end{array}$$

The previous lemma states that the top arrows are K-equivariantly homotopy equivalent via the coordinate flip. The vertical maps are chain-homotopy equivalent to the identity through the functor $\widehat{CC}((- \otimes \mathcal{A}) \rtimes K)$ via corollary 4.6. It ends up the proof. □

Proposition 4.18. *For any locally convex algebra A endowed with a smooth action of G, and of stabilization $\mathcal{A} = A \otimes_\pi \mathcal{D}$, the linear map $HP((\alpha^\sharp \otimes id_{\mathcal{A}}) \rtimes K)$ is left-invertible.*

Proof. Since the coordinates are independant to each others, the following diagram commutes

$$\begin{array}{ccc} q_1\mathbb{C} \otimes_\pi q_R\mathcal{M} & \xrightarrow{id_{q_1\mathbb{C}} \otimes \alpha^\sharp} & q_1\mathbb{C} \otimes_\pi M_2(\ell^{p,G}(\mathcal{H})) \\ {\scriptstyle \beta^\sharp \otimes id_{q_R\mathcal{M}}}\downarrow & & \downarrow{\scriptstyle \beta^\sharp \otimes id_{M_2}} \\ \mathcal{M} \otimes_\pi q_R\mathcal{M} & \xrightarrow[id_{\mathcal{M}} \otimes \alpha^\sharp]{} & \mathcal{M} \otimes_\pi M_2(\ell^{p,G}(\mathcal{H})) \end{array}$$

The theorem 4.5 states that the left-vertical arrow is nothing other than $\beta^\sharp \otimes id_{\mathcal{M}}$ through the functor $\widehat{CC}((- \otimes \mathcal{A}) \rtimes K)$. But through this functor, $\beta^\sharp \otimes id_{\mathcal{M}}$ is chain-homotopy equivalent to $id_{\mathcal{M}} \otimes \beta^\sharp$ by the previous proposition. It shows that

$$(\beta^\sharp \otimes id_{M_2}) \circ (id_{q_1\mathbb{C}} \otimes \alpha^\sharp) = (id_{\mathcal{M}} \otimes \alpha^\sharp) \circ (\beta^\sharp \otimes id_{q_R\mathcal{M}}) \sim_{\widehat{CC}((- \otimes \mathcal{A}) \rtimes K)} (id_{\mathcal{M}} \otimes \alpha^\sharp) \circ (id_{\mathcal{M}} \otimes \beta^\sharp).$$

Moreover, the second point of corollary 4.15 states that the last composition is the identity through the functor $HP_\bullet((- \otimes \mathcal{A}) \rtimes K)$. Namely, $HP((\beta^\sharp \otimes id_{M_2} \otimes id_{\mathcal{A}}) \rtimes K)$ realizes a left-inverse of $HP((id_{q_1\mathbb{C}} \otimes \alpha^\sharp \otimes id_{\mathcal{A}}) \rtimes K)$, which is chain-homotopy equivalent to $HP((\alpha^\sharp \otimes id_{\mathcal{A}}) \rtimes K)$ by theorem 4.5. □

Corollary 4.19. *Let G be an almost-connect real Lie group and K be a maximal compact subgroup of it. For every unital locally convex algebra A endowed with a smooth action of G, the Dirac element $\alpha \in KK_G(\mathcal{C}_\tau(G/K), \mathbb{C})$ induces an isomorphism in periodic cyclic homology*

$$(\alpha^\sharp \otimes id_{\mathcal{A}}) \rtimes G : HP_\bullet(\mathcal{C}_c^\infty(G/K, \mathcal{A}) \rtimes G) \xrightarrow{\sim} HP_{\bullet+\dim(G/K)}(\mathcal{A} \rtimes G)$$

where $\mathcal{A} = A \otimes_\pi \mathcal{D}$ is the stabilized algebra.

Proof. When G/K is even-dimensional, the corollary 4.15 and the proposition 4.18 together realize the hypothesis of the strategy 3.3 which makes isomorphism occur in this case. The odd-dimensional case being a consequence of the lemma 3.1. □

Theorem 4.20 (Main theorem)**.** *Let G be an almost-connect real Lie group and K be a maximal compact subgroup of it. For every unital locally convex algebra A endowed with a smooth action of G, the Dirac element $\alpha \in KK_G(\mathcal{C}_\tau(G/K), \mathbb{C})$ induces an isomorphism in periodic cyclic homology*

$$HP((\alpha^\sharp \otimes id_{\mathcal{A}}) \rtimes G) : HP_\bullet(\mathcal{A} \rtimes K) \simeq HP_\bullet(\mathcal{C}_c^\infty(G/K, \mathcal{A}) \rtimes G) \xrightarrow{\sim} HP_{\bullet+\dim(G/K)}(\mathcal{A} \rtimes G)$$

where $\mathcal{A} = A \otimes_\pi \mathcal{D}$ is the stabilized algebra.

Proof. Follows from 4.19 and 2.13 applied to $H = K$. □

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